2026-06-03

Vision AI Already Knows:

Understanding the World Without Labels

Sanghyun Jo

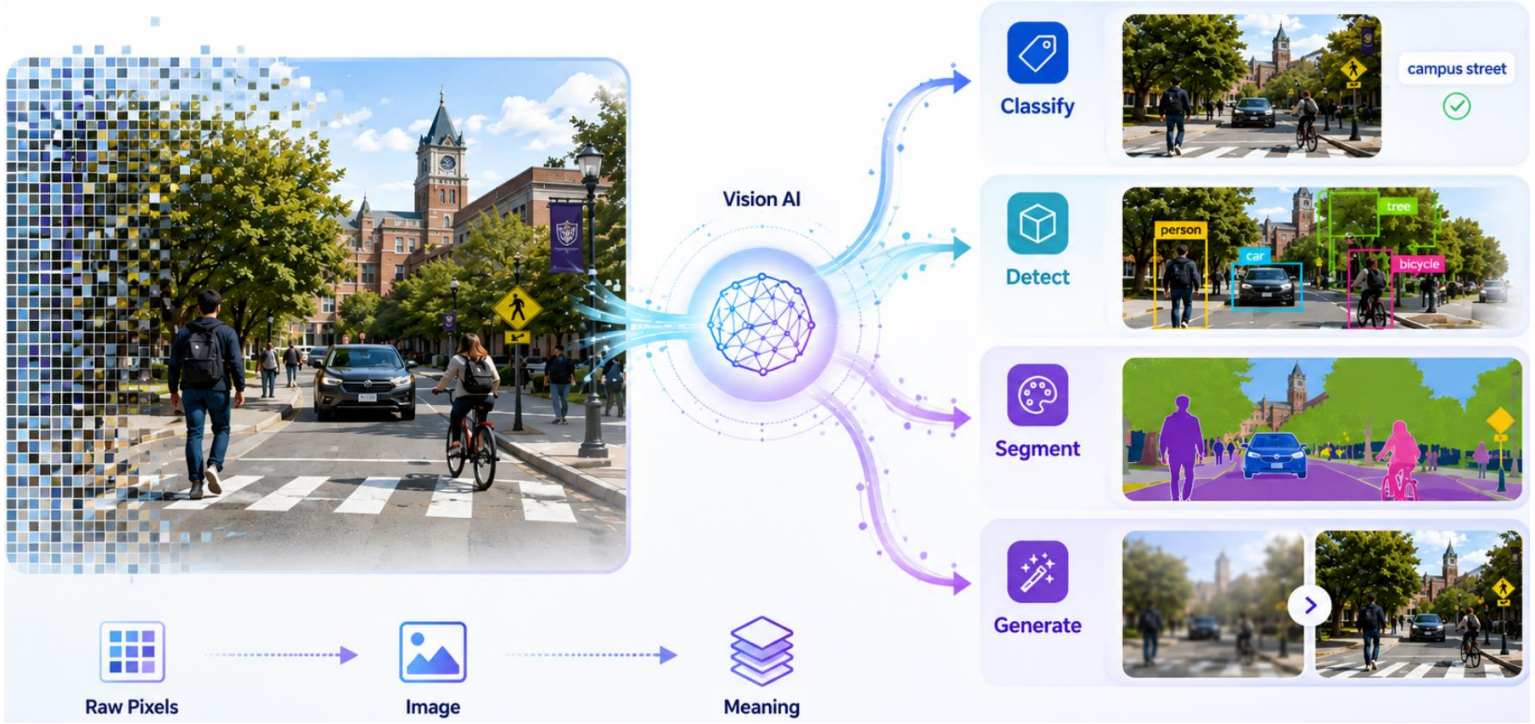
Principal AI Researcher, OGQ

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Homepage

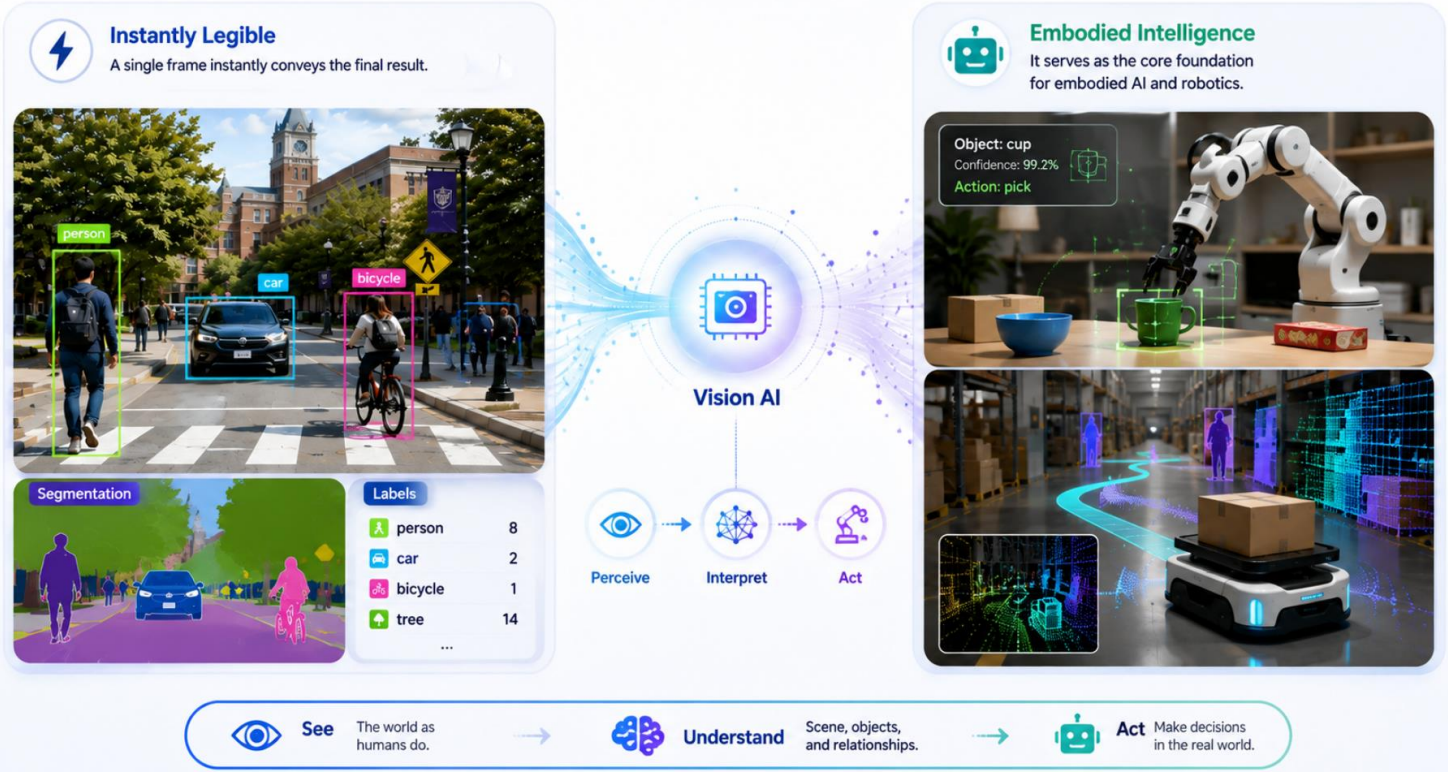
Decoding the Visual World

- **Translating Raw Data:** Vision AI extracts structured meaning from raw pixels through classification, detection, segmentation, and generation.
- **Intuitive Intelligence:** The before/after images speak for themselves.



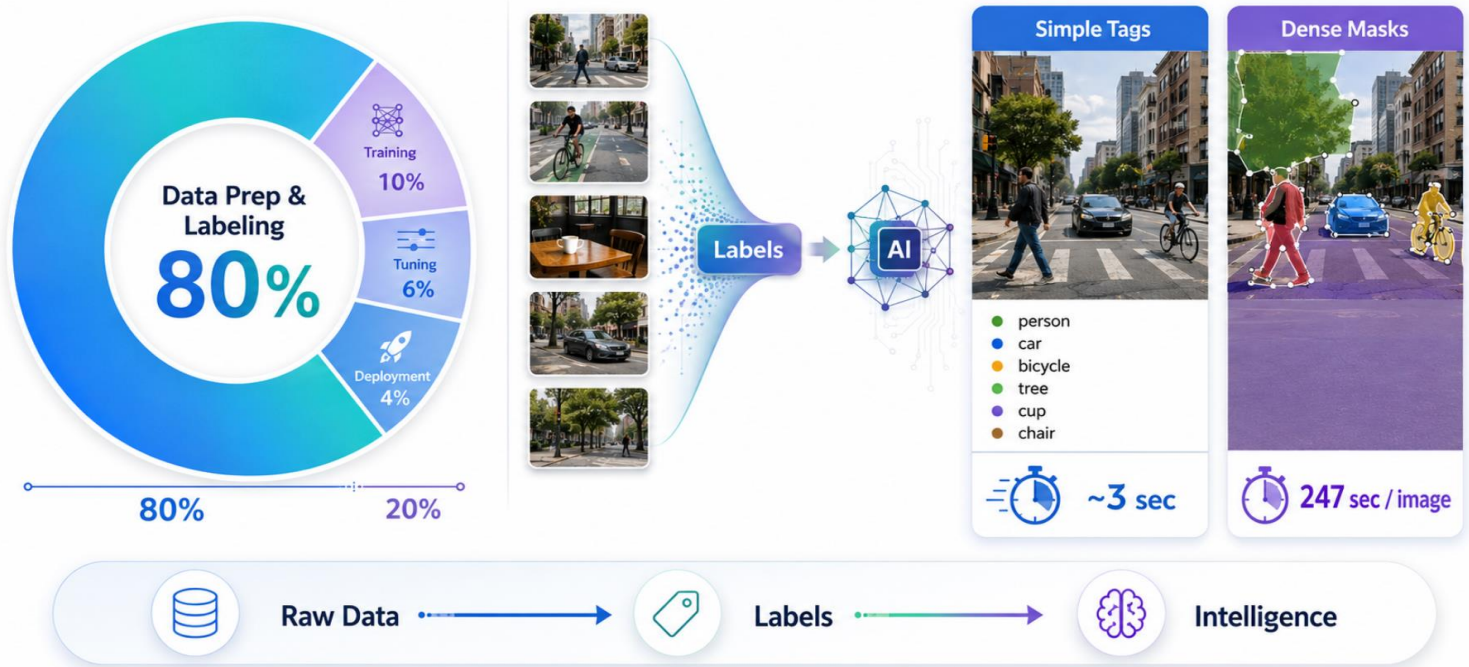
Seeing is Believing

- **Immediate Impact:** A single frame instantly conveys the result.
- **Long Term Vision:** It serves as the core foundation for embodied AI and robotics.



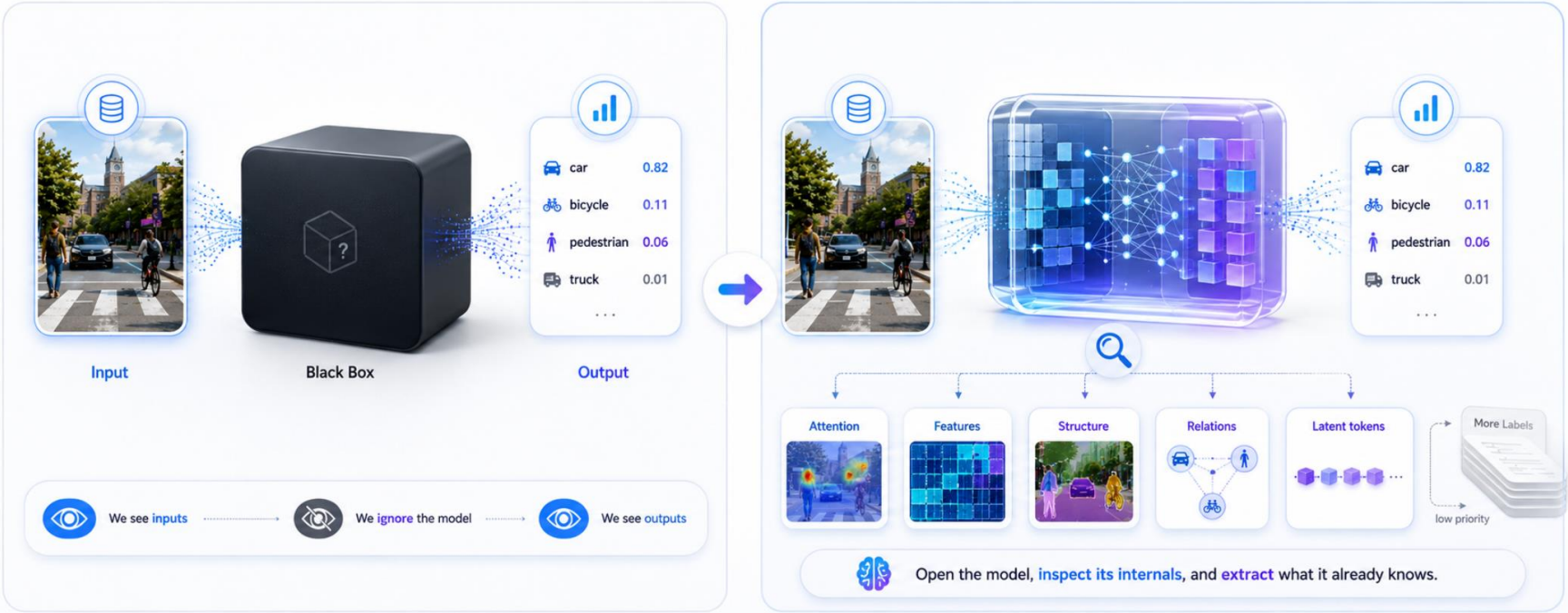
The 80 Percent Trap: The Hidden Tax of AI

- **The Hidden Cost:** Data cleansing and labeling consume 80 percent of the entire AI workflow.
- **The Annotation Penalty:** Pixel-level masking takes ~247 seconds/image, while tags only take seconds.



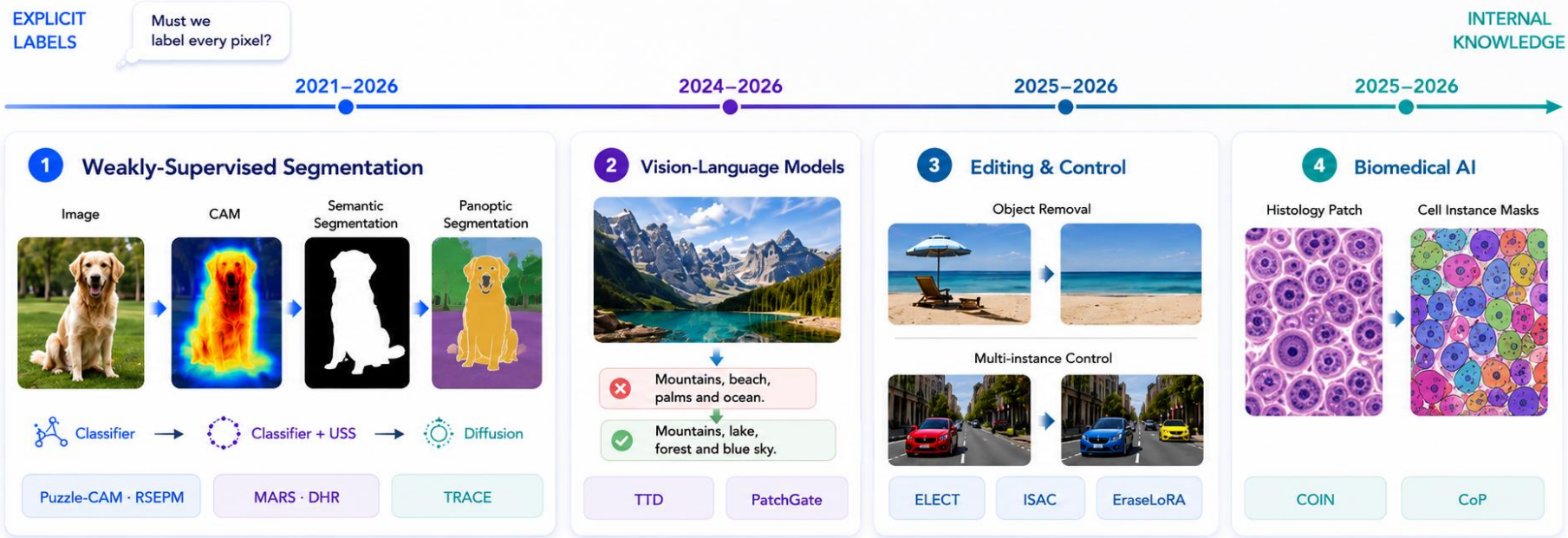
Shattering the Black Box

- **The Status Quo:** The industry focuses on inputs and outputs while ignoring internal mechanisms.
- **The Paradigm Shift:** Instead of blindly collecting more labels, we must uncover what the model already knows.



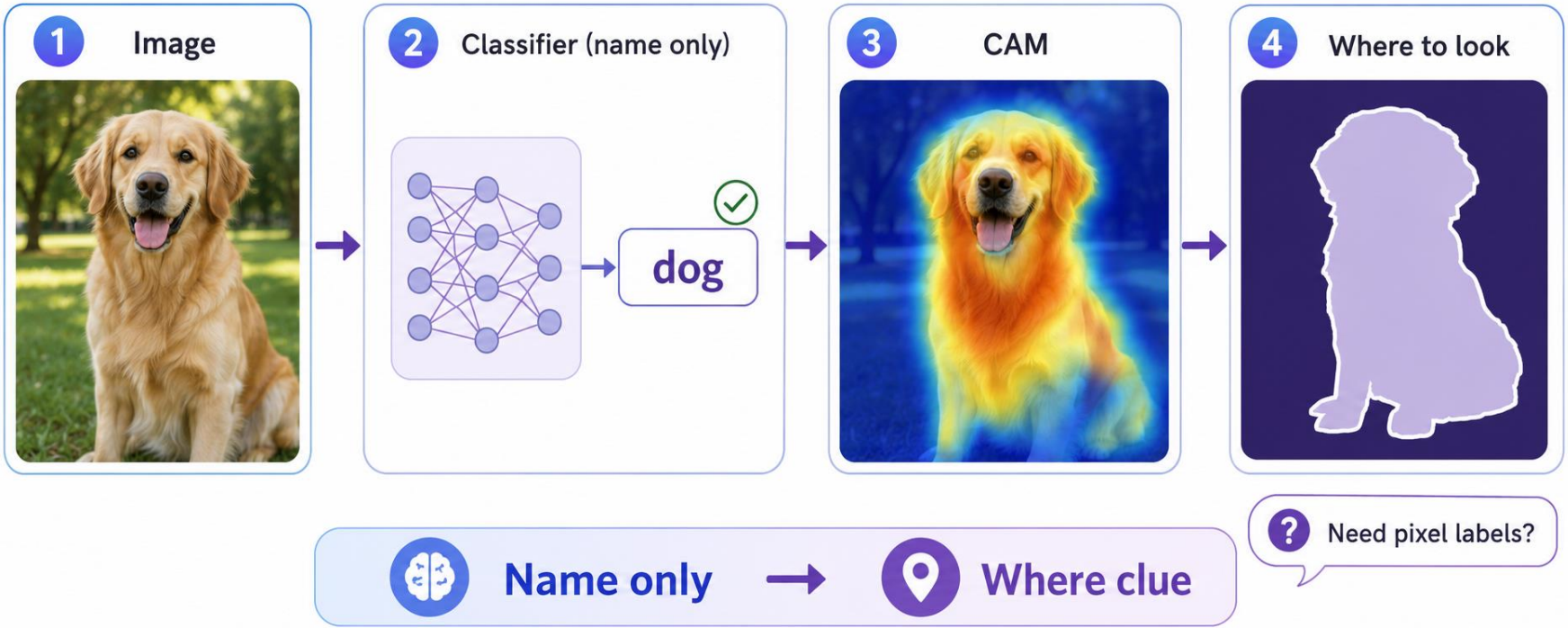
A Five Year Journey: Defying the Label

- **A Simple Question:** Must we label every pixel? This has driven my research from 2021 to today.
- **Evolving Targets:** Expanding from image classifiers to multimodal and diffusion models.



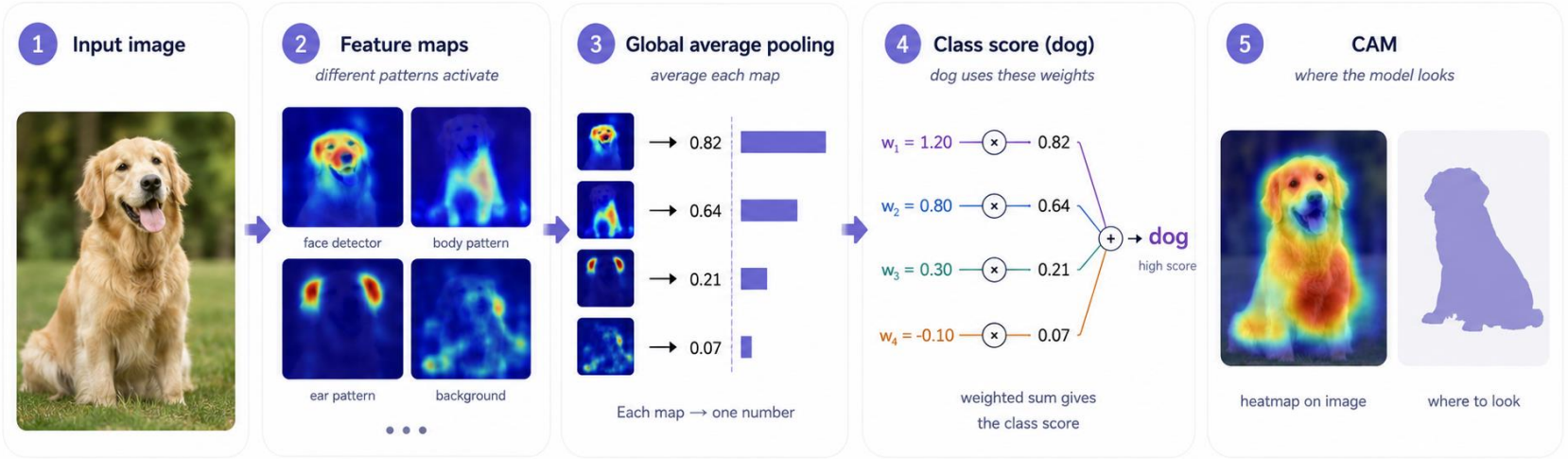
The Classifier Already Knows Where to Look

- A classifier trained only to name an image reveals where the object is through its Class Activation Map.
- The seed of everything. If the model already points at the object, do we still need pixel labels?



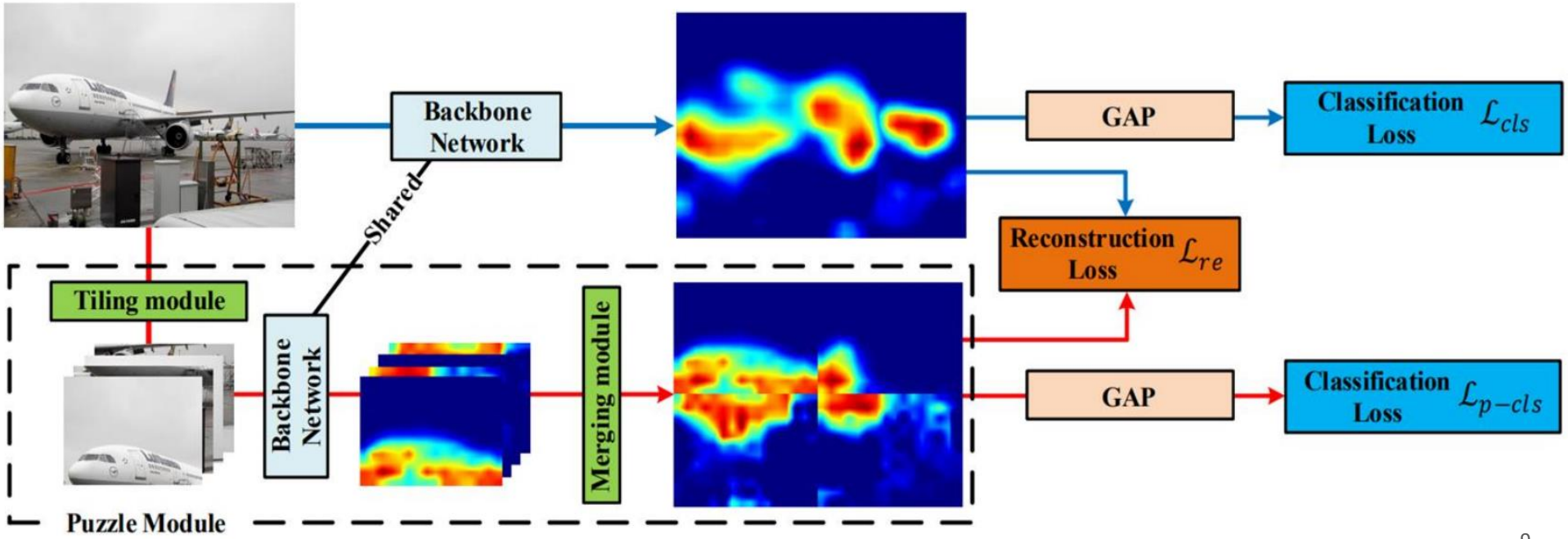
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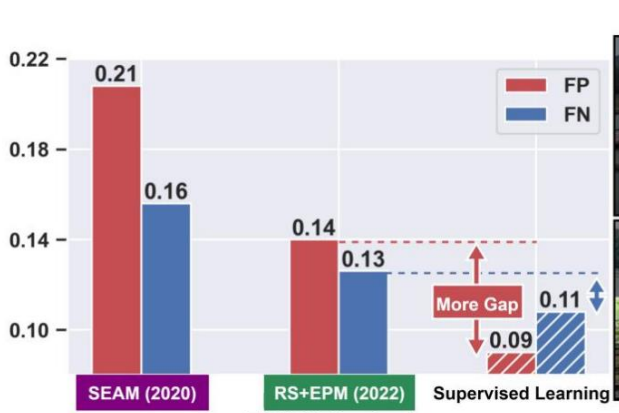
[ICIP 2021] Puzzle-CAM: Seeing the Whole, Not Just the Part

- Tiling the image and matching partial against full features expands coverage beyond the most discriminative part.
- Achieves dense localization from image level tags alone.

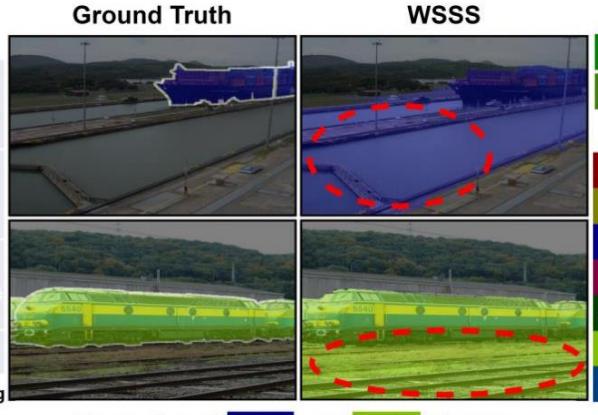


[ICCV 2023] MARS: Removing Hidden Bias Automatically

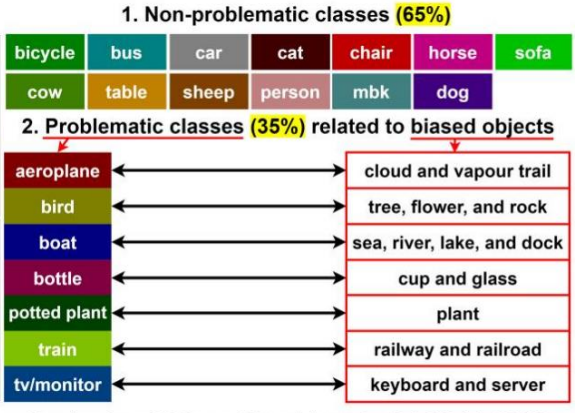
- Classifiers learn false shortcuts, such as associating a boat with water, affecting ~35 percent of VOC.



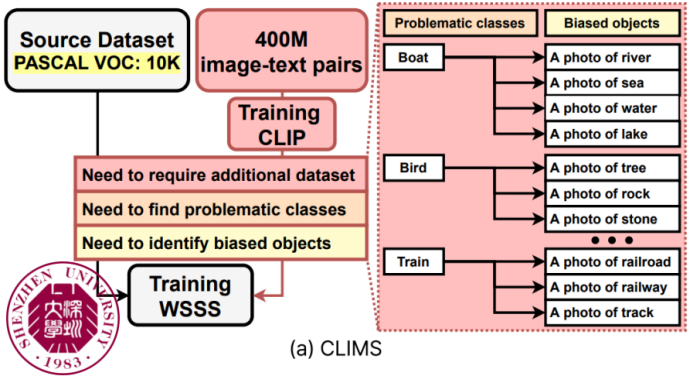
FP/FN Analysis



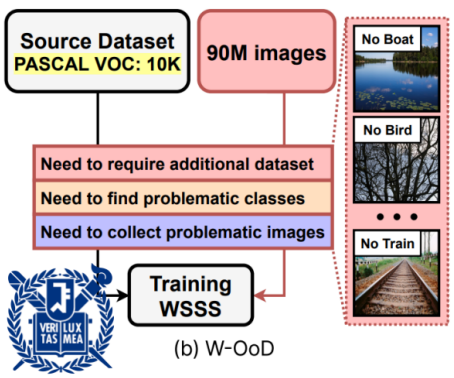
Examples of boat and train classes



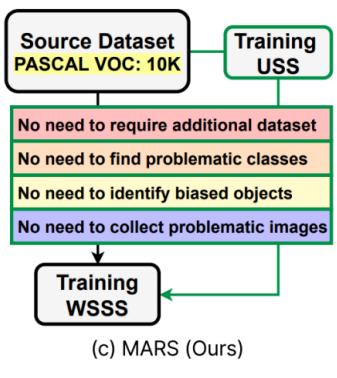
Analysis of Biased Problem in PASCAL VOC



(a) CLIMS



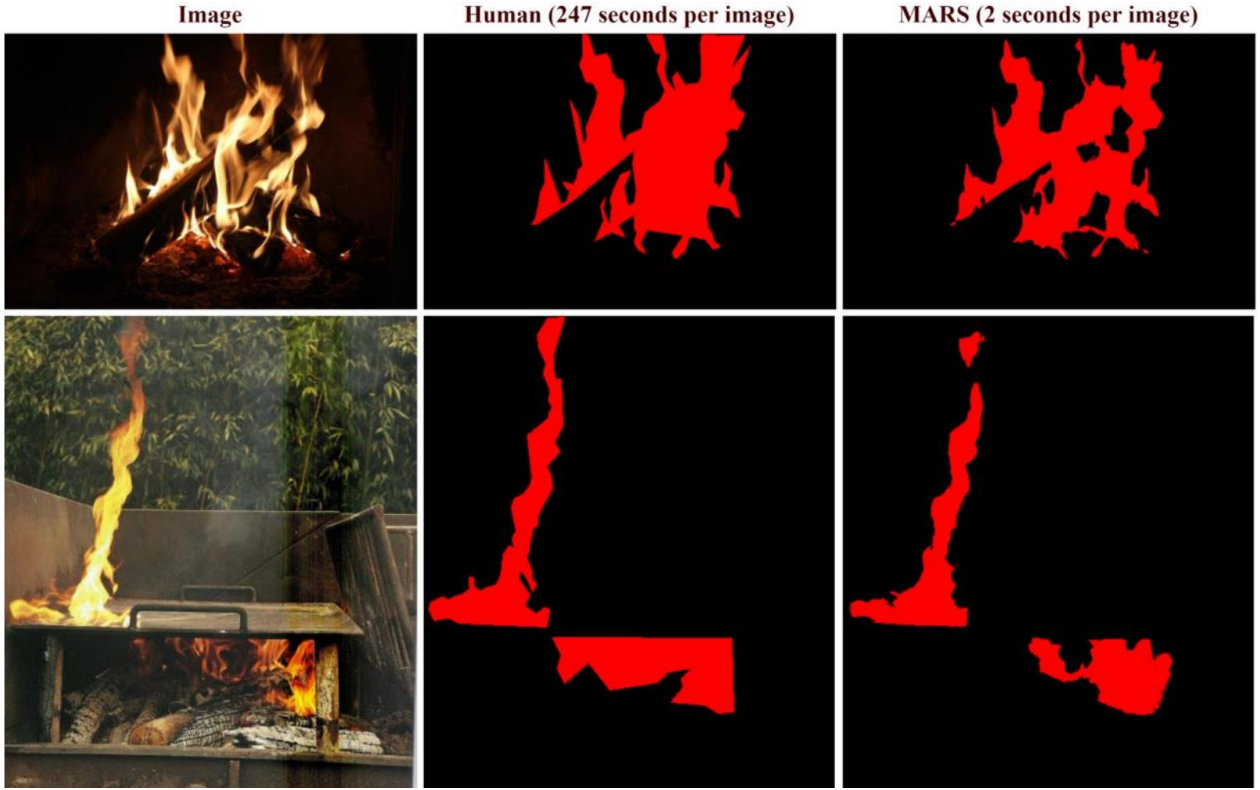
(b) W-OoD



(c) MARS (Ours)

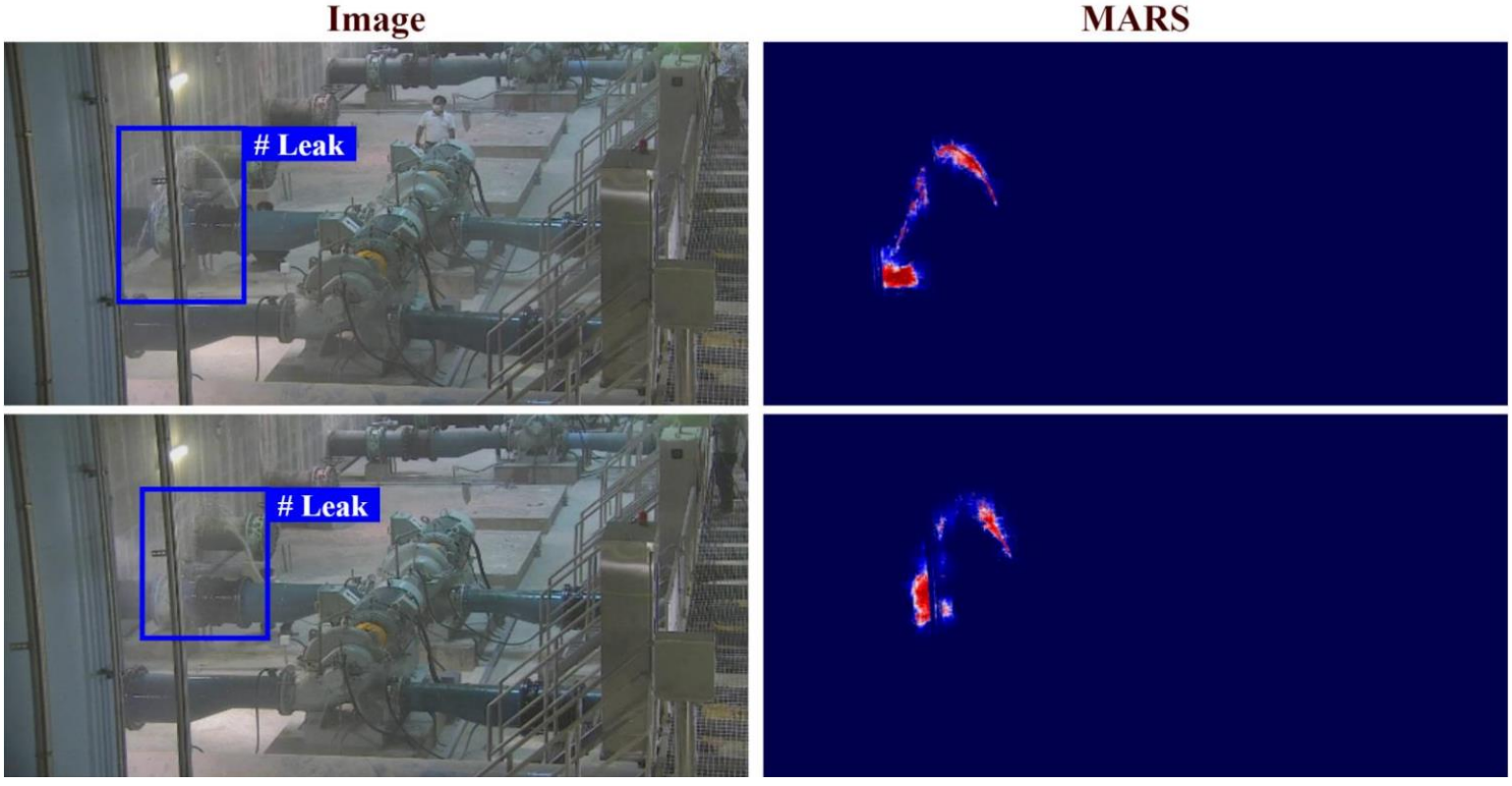
[ICCV 2023] MARS: Removing Hidden Bias Automatically

- Removed with zero extra supervision, reaching SOTA and cutting labeling time from 247 seconds to 2 seconds per image.



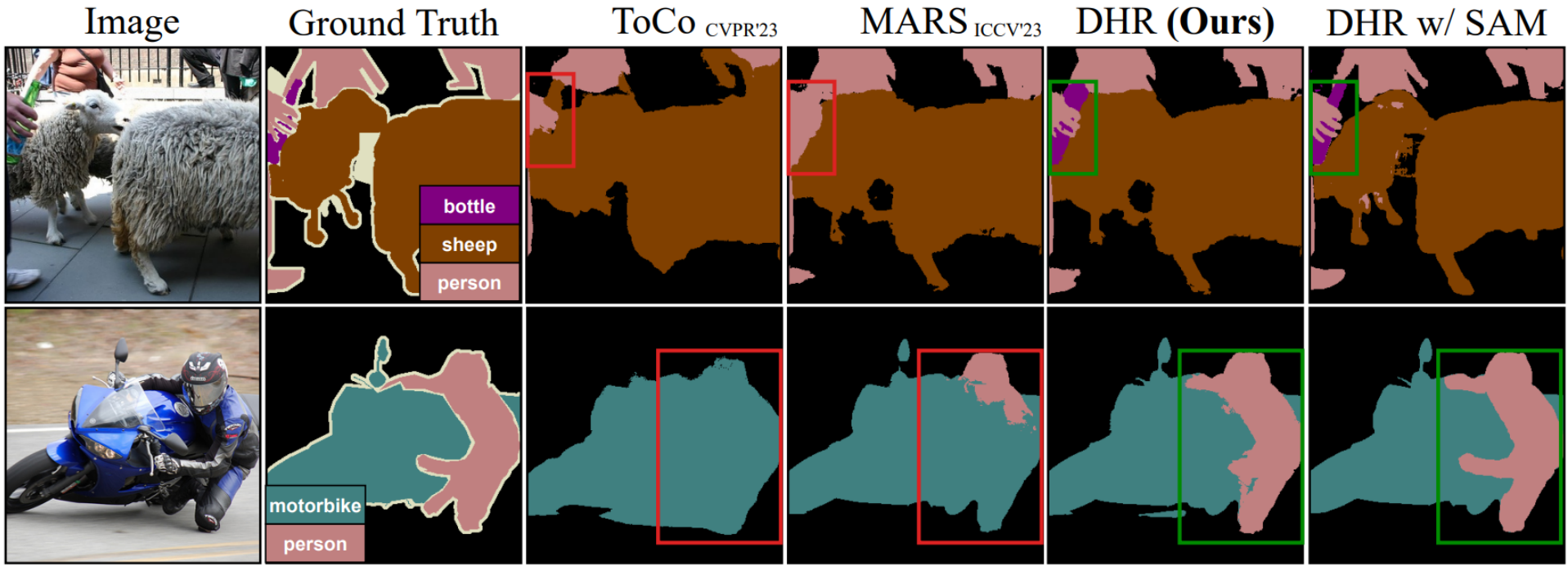
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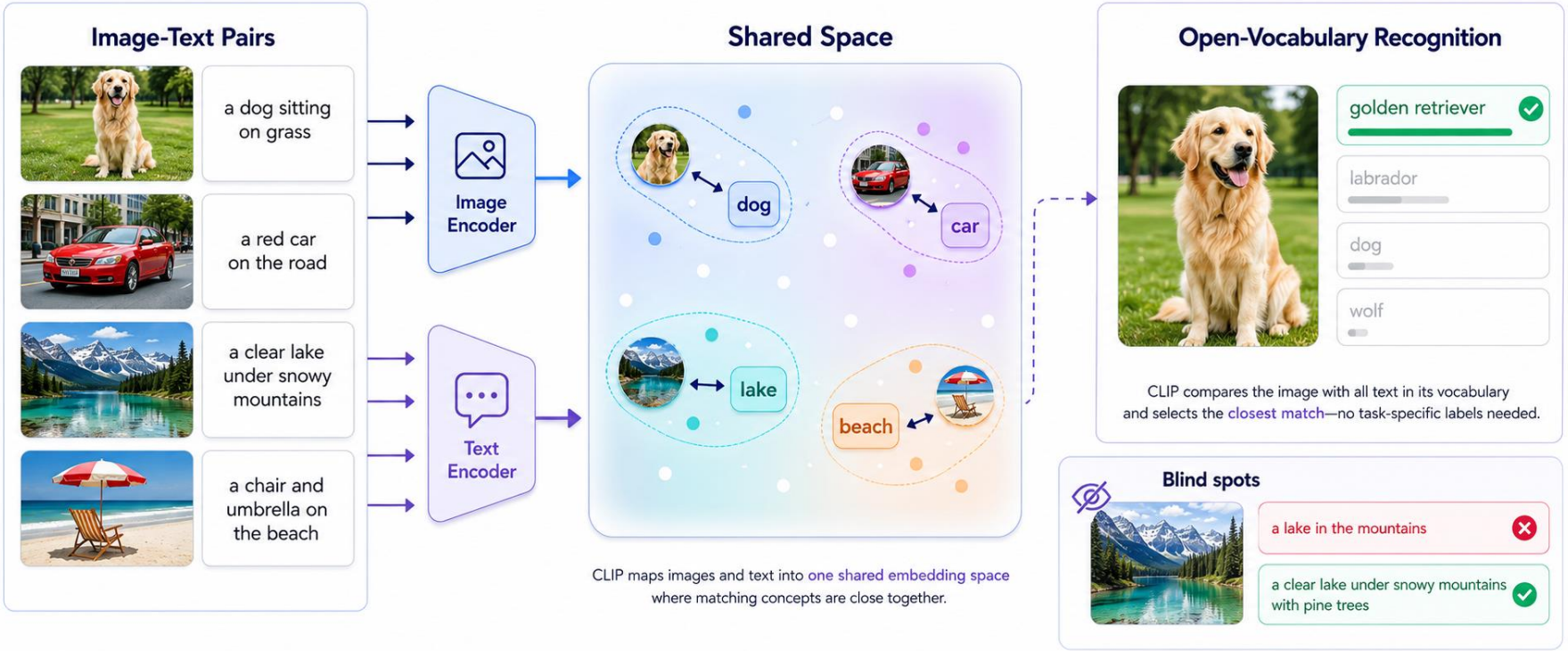
[ECCV 2024] DHR: Balancing What the Model Overlooks

- Combining two feature sources rebalances underrepresented inter- and intra-class regions.
- Rare and small regions stop disappearing, significantly lifting segmentation quality.



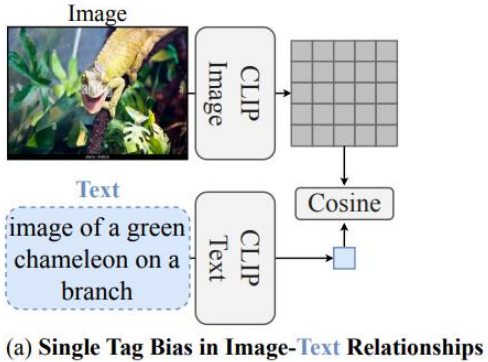
What is a Vision-Language Model (VLM)?

- CLIP learns from millions of image-text pairs, placing pictures and words into one shared space.
- This enables zero-shot classification w/o task-specific labels, though it inherits blind spots that require fixing.

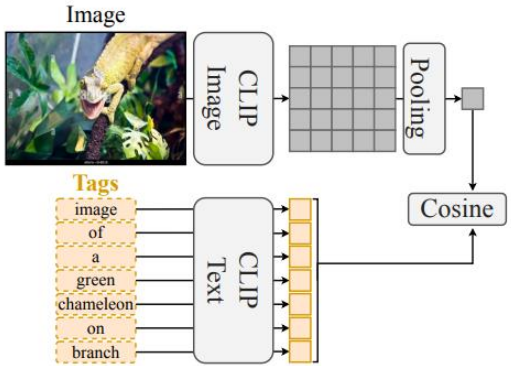


[ECCV 2024] TTD: Fixing the Single-Tag Blind Spot

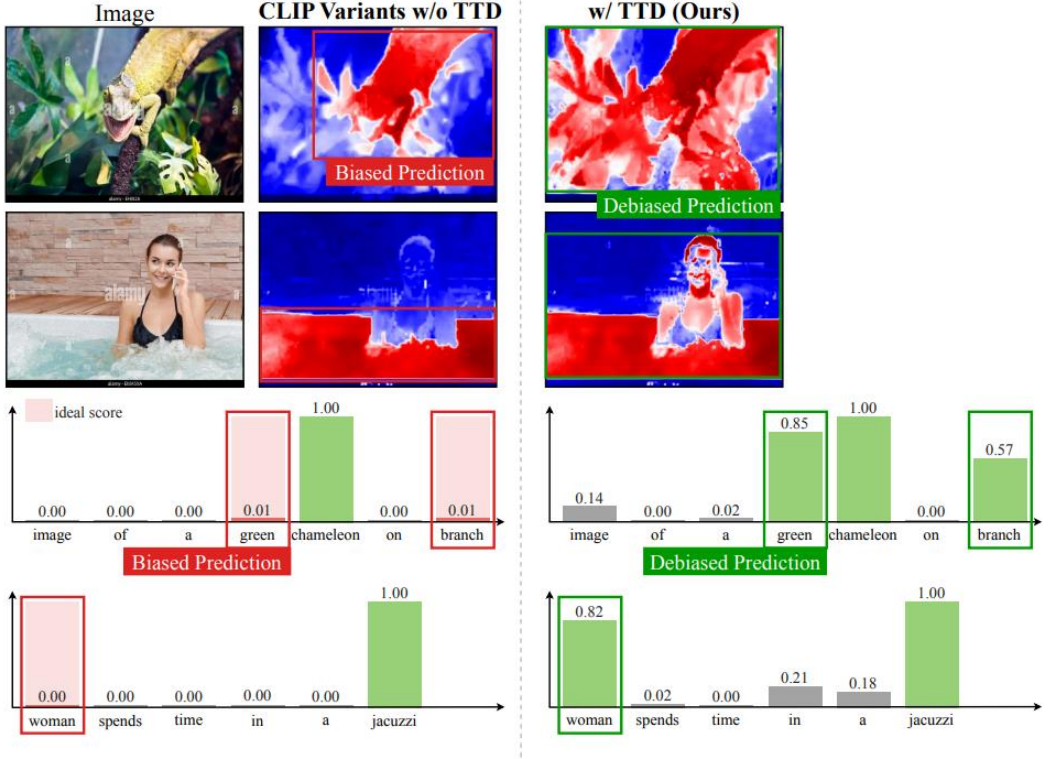
- CLIP tends to latch onto one dominant tag and ignore the rest.
- TTD self-distills text tags to restore full image-text alignment, improving multi-label understanding.



(a) **Single Tag Bias in Image-Text Relationships**

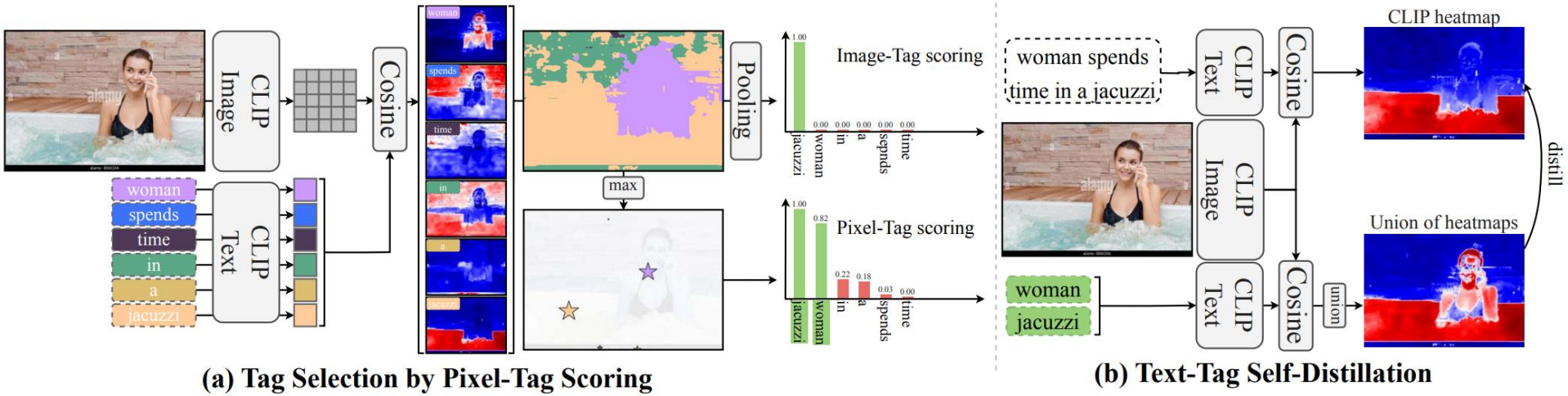


(b) **Single Tag Bias in Image-Tags Relationships**



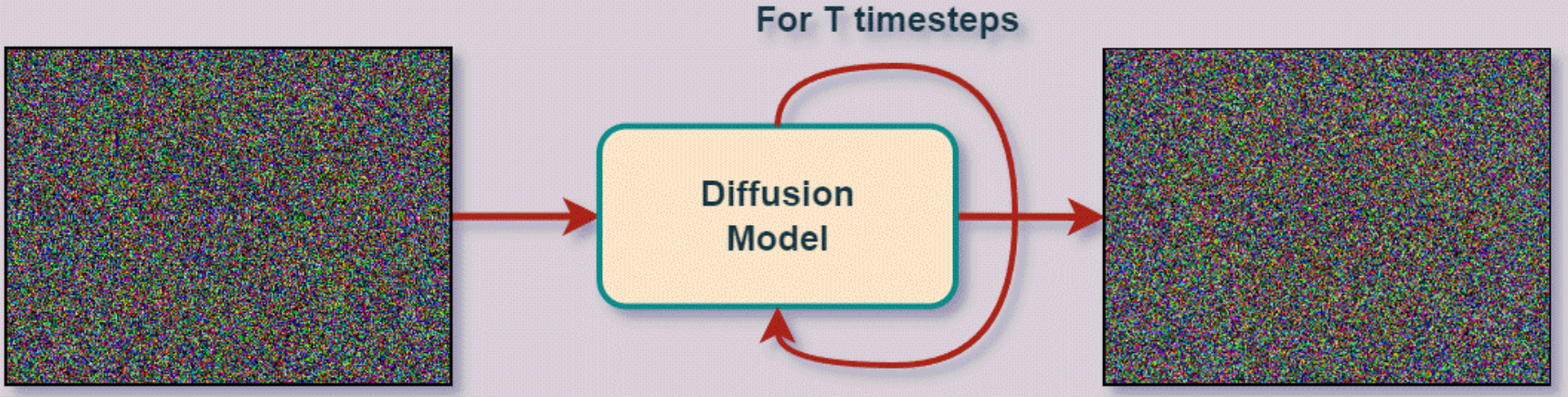
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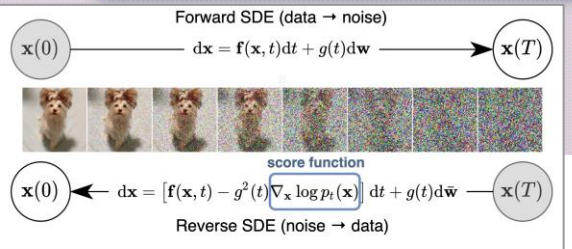
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What is a Diffusion Model?

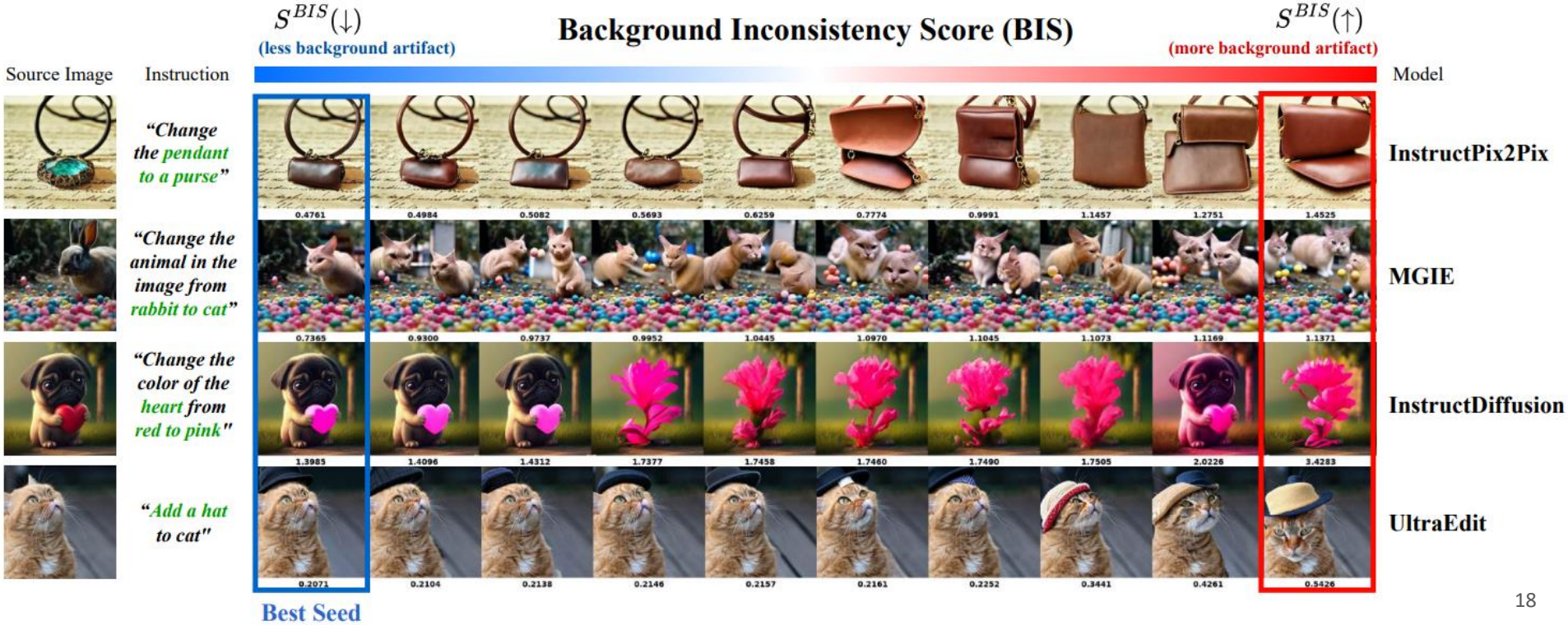
- A diffusion model turns pure noise into an image by removing noise step-by-step.
- The reverse process is not just for drawing. Its intermediate steps hold structures we can read & steer.





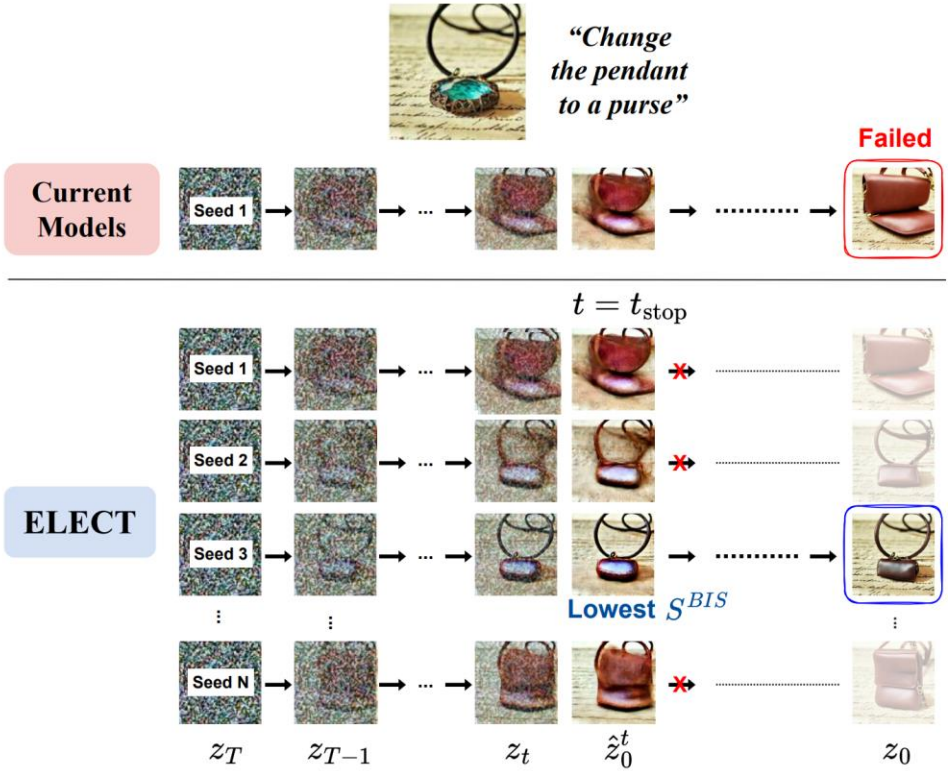
[ICCV 2025] ELECT: Choosing the Right Edit Early

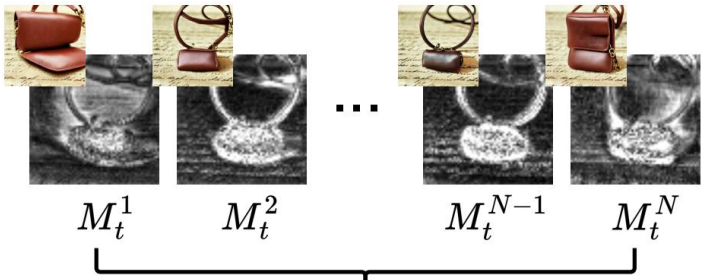
- For instruction-guided editing, ELECT picks the most promising candidate at an early denoising step.
- Zero-shot and no training required, saving significant compute on dead-end edits.



[ICCV 2025] ELECT: Choosing the Right Edit Early

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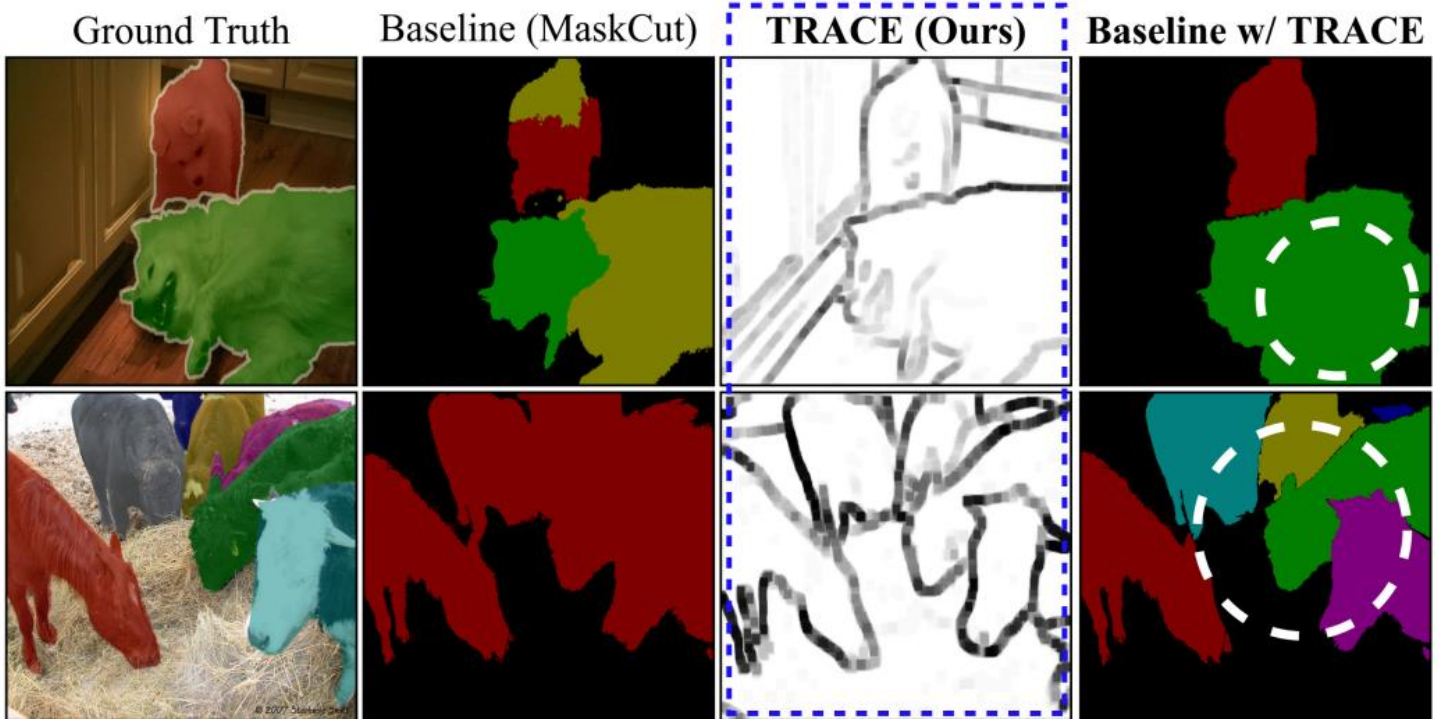
$$M_t^i = |\epsilon_{cond} - \epsilon_{uncond}|$$
$$M_t^{mean} = \frac{1}{|\mathbb{S}|} \sum_{i \in \mathbb{S}} M_t^i$$



$$S^{\text{BIS}}(i,t) := (1 - (M_t^{\text{mean}})^2) \odot |\hat{z}_0^i - \mathcal{E}(I)|$$

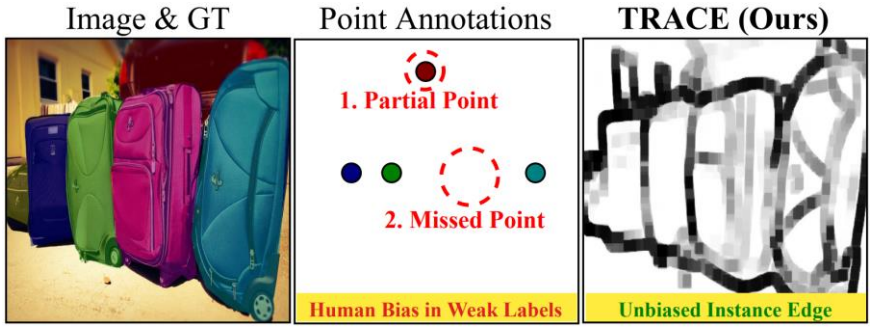
[ICLR 2026 Oral] TRACE: The Secret Life of Diffusion Models

- Extracts pure, instance-aware boundaries instead of relying on semantic clustering.
- Uses these edges as barriers to separate adjacent objects for merging and fragmentation.



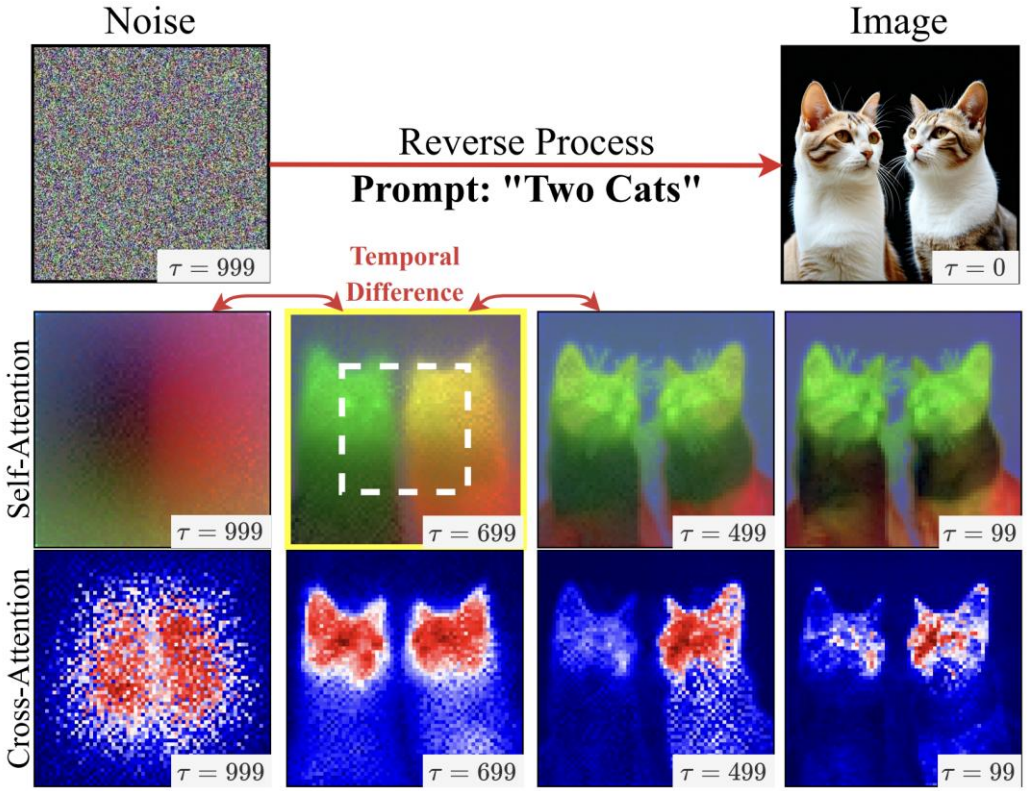
[ICLR 2026 Oral] TRACE: The Secret Life of Diffusion Models

Method	Backbone	Supervision	VOC 2012					COCO 2017				
			PQ	PQ^{th}	PQ^{st}	SQ	RQ	PQ	PQ^{th}	PQ^{st}	SQ	RQ
Panoptic FCN (Li et al., 2021)	ResNet-50	$\mathcal{M}$	67.9	66.6	92.9	-	-	43.6	49.3	35.0	80.6	52.6
Mask2Former* (Cheng et al., 2022)	ResNet-50	$\mathcal{M}$	73.6	72.6	93.5	90.6	80.5	51.9	57.7	43.0	-	-
PSPS (Fan et al., 2022)	ResNet-50	$\mathcal{P}$	49.8	47.8	89.5	-	-	29.3	29.3	29.4	-	-
Panoptic FCN (Li et al., 2021)	ResNet-50	$\mathcal{P}$	48.0	46.2	85.2	-	-	31.2	35.7	24.3	-	-
Point2Mask* (Li et al., 2023b)	ResNet-50	$\mathcal{P}$	53.8	51.9	90.5	-	-	32.4	32.6	32.2	75.1	41.5
EPLD (Li et al., 2024)	ResNet-50	$\mathcal{P}$	56.6	54.9	89.6	-	-	34.2	33.6	35.3	-	-
Point2Mask* (Li et al., 2023b)	Swin-L	$\mathcal{P}$	61.0	59.4	93.0	-	-	37.0	37.0	36.9	75.8	47.2
EPLD (Li et al., 2024)	Swin-L	$\mathcal{P}$	68.5	67.3	93.4	-	-	41.0	39.9	42.7	-	-
JTSM (Shen et al., 2021)	ResNet-18-WS	$\mathcal{I}$	39.0	37.1	77.7	-	-	5.3	8.4	0.7	30.8	7.8
MARS* (Jo et al., 2023)	ResNet-50	$\mathcal{I}$	41.4	39.8	85.3	83.0	57.8	11.7	13.3	10.2	58.3	11.8
+ TRACE (Ours)	ResNet-50	$\mathcal{I}$	50.4	48.5	88.9	86.6	60.1	29.5	31.1	28.9	62.5	39.3
DHR* (Jo et al., 2024a)	ResNet-50	$\mathcal{I}$	45.0	43.3	88.3	83.3	59.8	18.3	17.5	18.1	69.3	14.8
+ TRACE (Ours)	ResNet-50	$\mathcal{I}$	56.9	55.2	91.0	88.4	63.4	32.8	32.7	32.9	75.5	42.5
+ TRACE (Ours)	Swin-L	$\mathcal{I}$	**69.8**	**68.4**	**96.2**	**94.5**	**71.2**	**43.1**	**42.5**	**43.5**	**83.8**	**55.3**



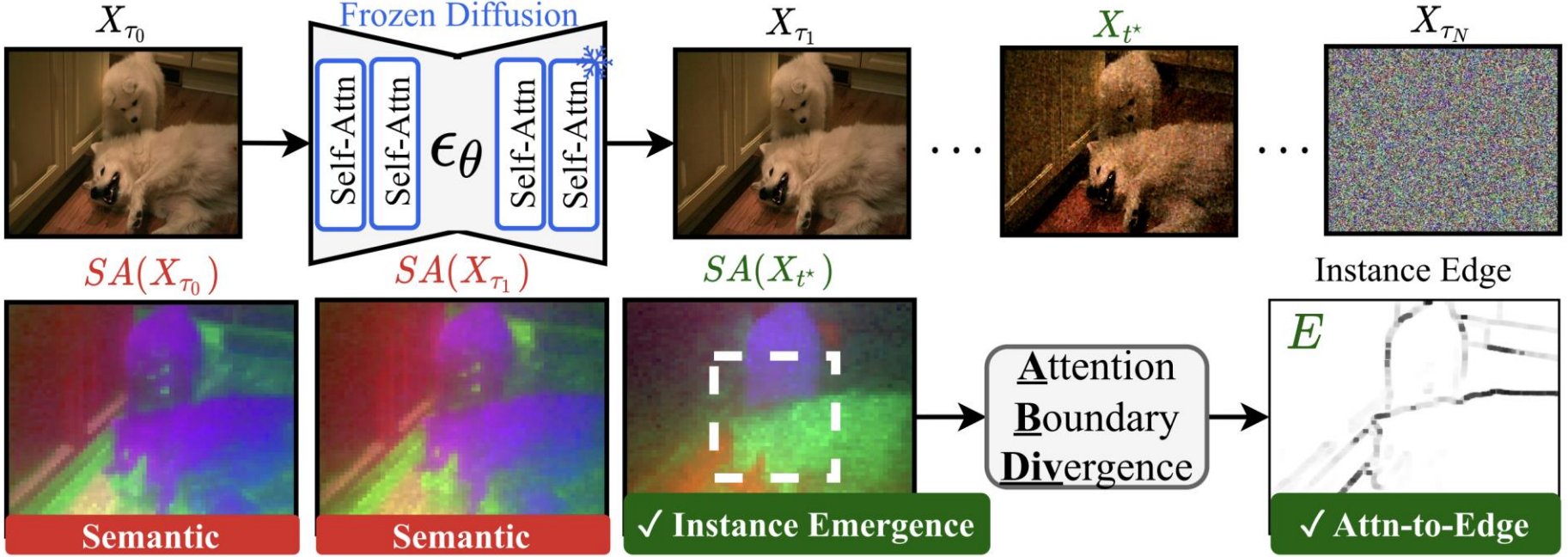
[ICLR 2026 Oral] TRACE: The Secret Life of Diffusion Models

- Our hidden prior reveals that diffusion models naturally separate objects on their own during denoising steps.
- Boundary-centric separation extracts precise instance edges with zero human annotations.



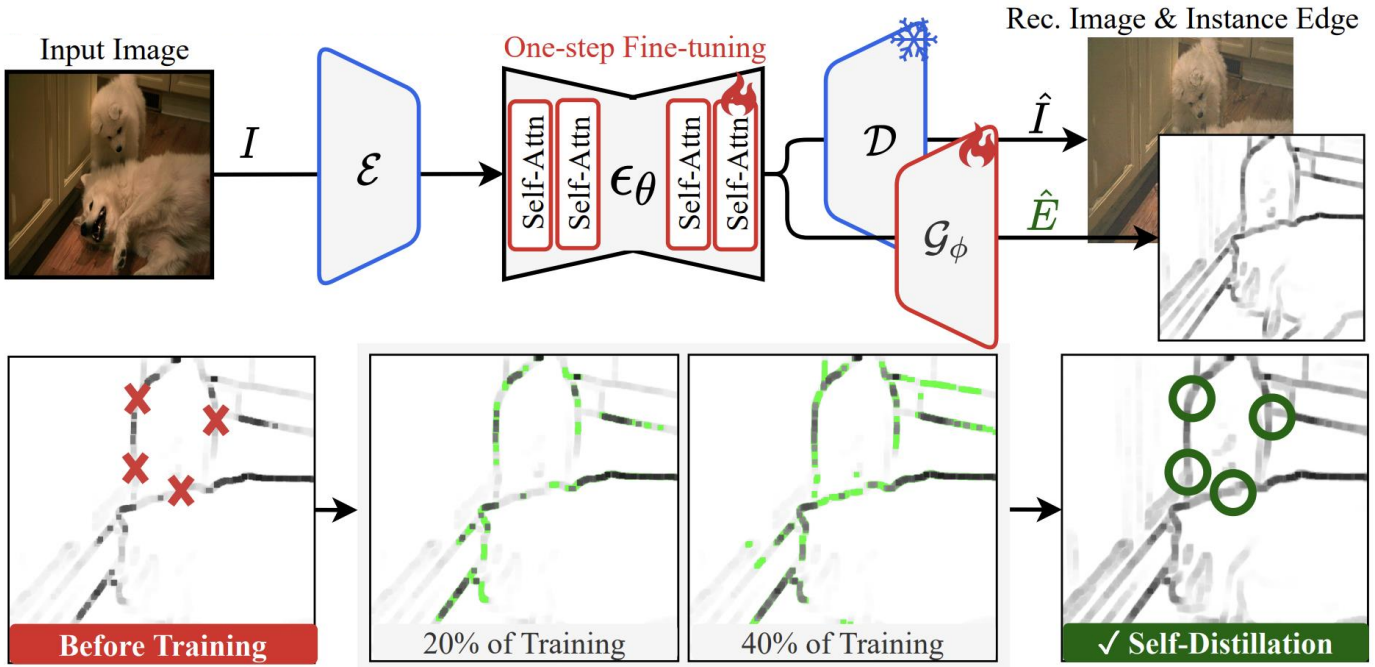
[ICLR 2026 Oral] TRACE: The Secret Life of Diffusion Models

- **IEP:** Tracks the forward process to pinpoint t*, the exact timestep where self-attention shifts from semantic blobs to sharp instance structures.
- **ABDiv:** Transforms this attention map into an initial edge map w/o requiring any annotations.



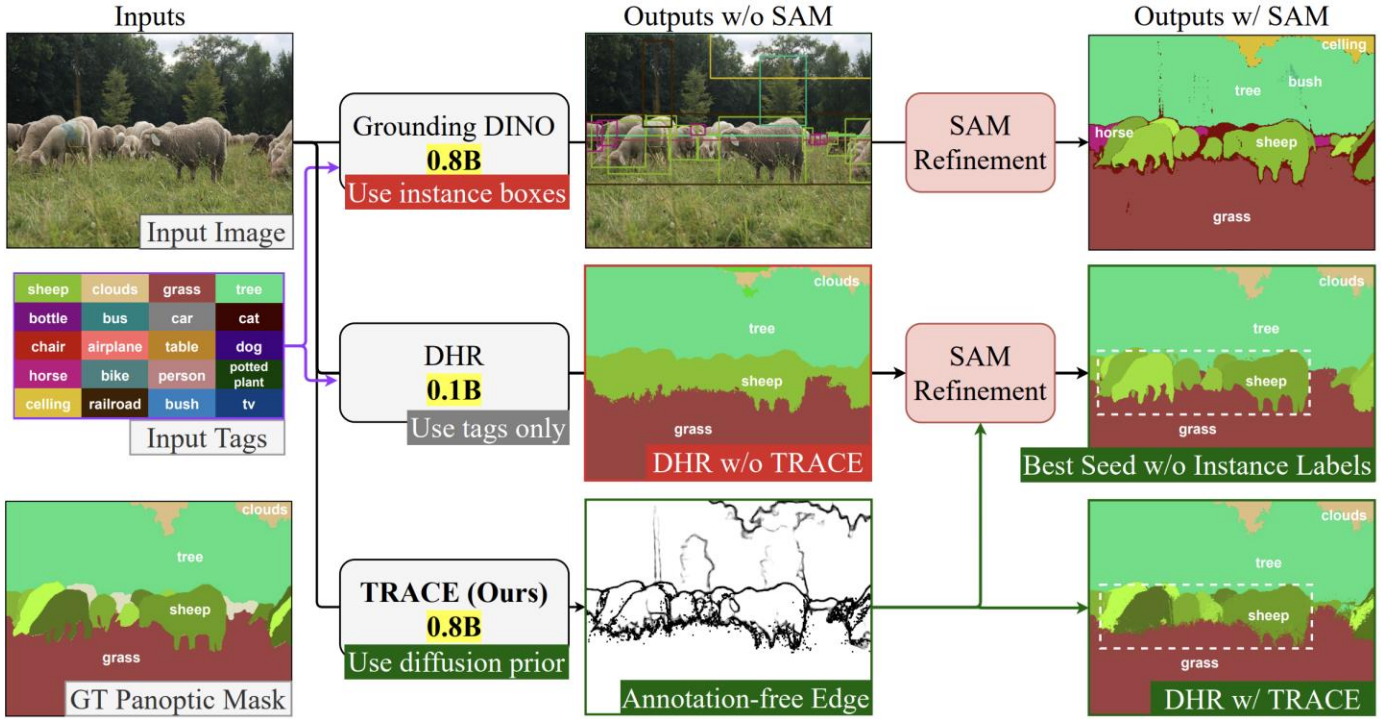
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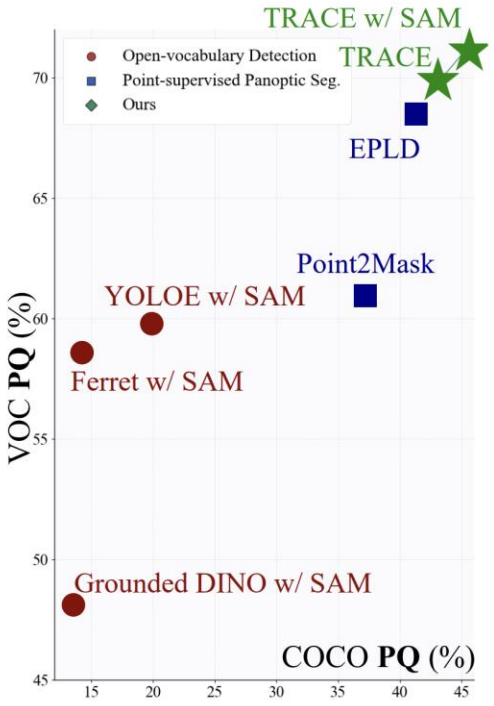
- **Self-Distillation:** Fine-tunes the diffusion backbone (via LoRA) alongside a lightweight edge decoder, using Stage 1 pseudo-edges as training targets.
- **Real-Time Inference:** Eliminates the iterative IEP search, slashing inference latency from over 3 seconds to just 45ms per image (an 81x speedup).



[ICLR 2026 Oral] TRACE: The Secret Life of Diffusion Models

- **Annotation-Free Seeds:** Converts tag-supervised masks into panoptic masks.
- **Surpassing SOTA:** Outperforms point- and box-supervised methods w/o instance labels

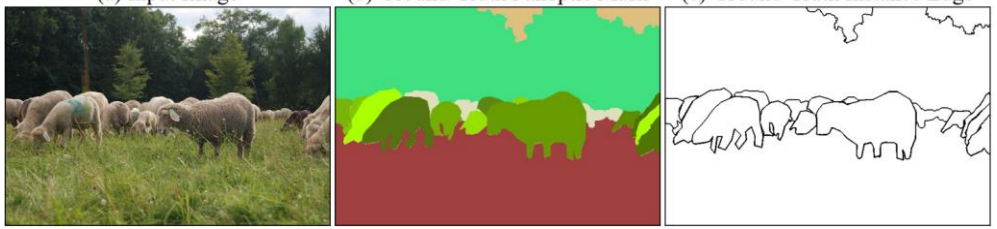




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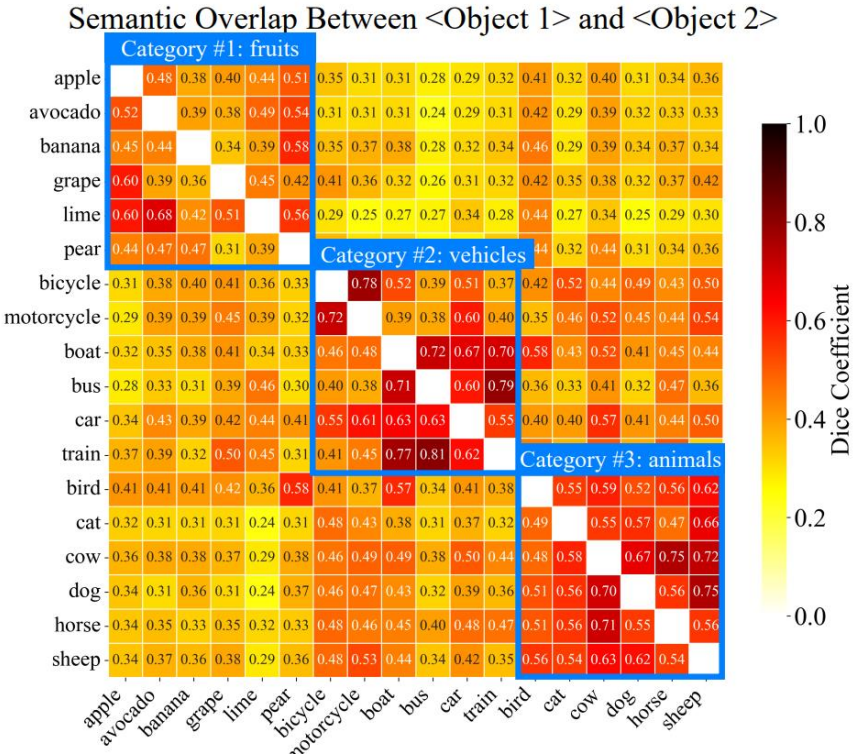
Method	Backbone	Params	AP^{mk}	AR^{mk}_{100}
ProMerge	–	–	3.1	7.6
Non-diffusion backbones (ABDiv only)				
+ TRACE	DINOv2-G	1.1B	2.6	7.7
+ TRACE	EVA02-E	5.0B	3.2	7.9
+ TRACE	DINOv3	7.0B	4.3	8.9
+ TRACE	LLaVA	13B	3.8	8.4
+ TRACE	Qwen2.5-VL	72B	4.1	8.5
Diffusion backbones (IEP + ABDiv)				
+ TRACE	SD1.5	0.8B	6.8	11.2
+ TRACE	PixArt-α	**0.6B**	7.1	11.8
+ TRACE	SDXL	2.5B	7.4	12.3
+ TRACE	SD3.5-L	8.1B	8.2	13.1
+ TRACE	FLUX.1	12B	**8.3**	**13.4**

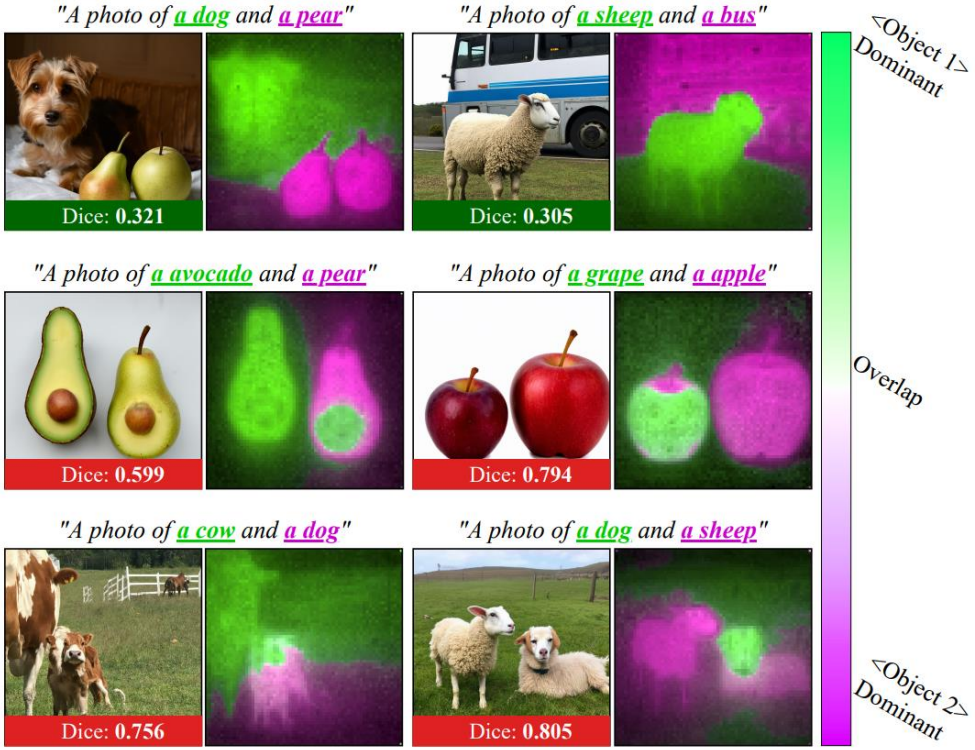


Method	ODS	OIS	clDice
Canny	0.129	0.202	0.134
HED	0.347	0.443	0.446
PiDiNet	0.362	0.450	0.574
DiffusionEdge	0.428	0.485	0.576
TRACE (Ours)	**0.889**	**0.899**	**0.826**

[Under Review] ISAC: Making Diffusion Count Correctly

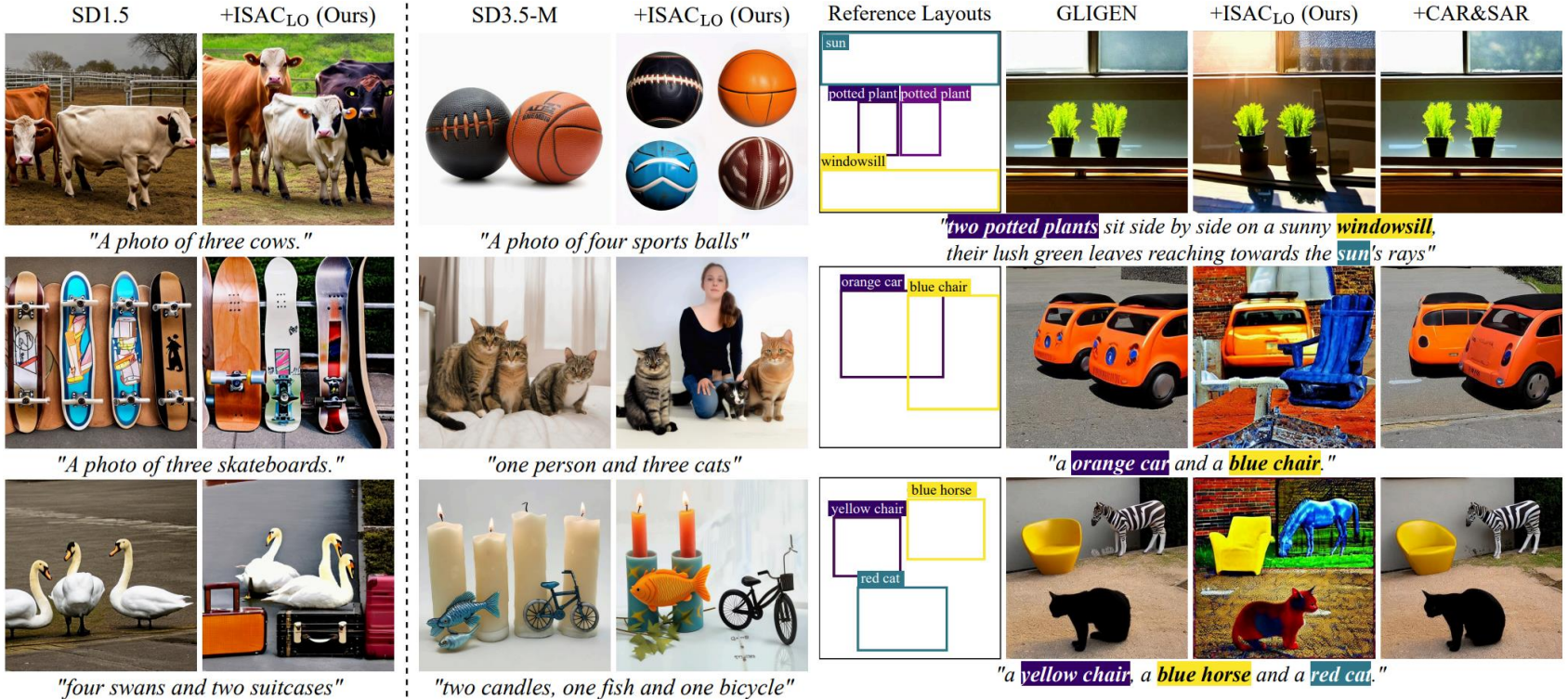
- Text-to-image diffusion often blurs multiple instances together.





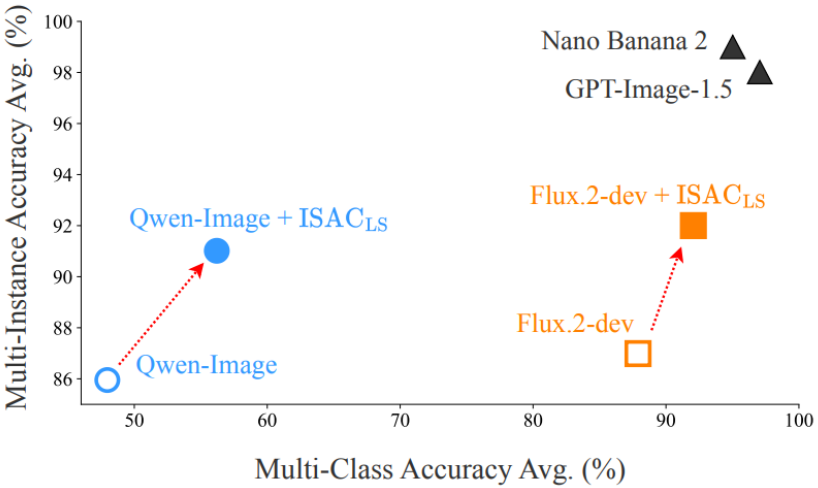
[Under Review] ISAC: Making Diffusion Count Correctly

- ISAC steers the model's internal attention from instances to semantics so each object renders distinctly.
- Training-free control for reliable multi-instance generation.

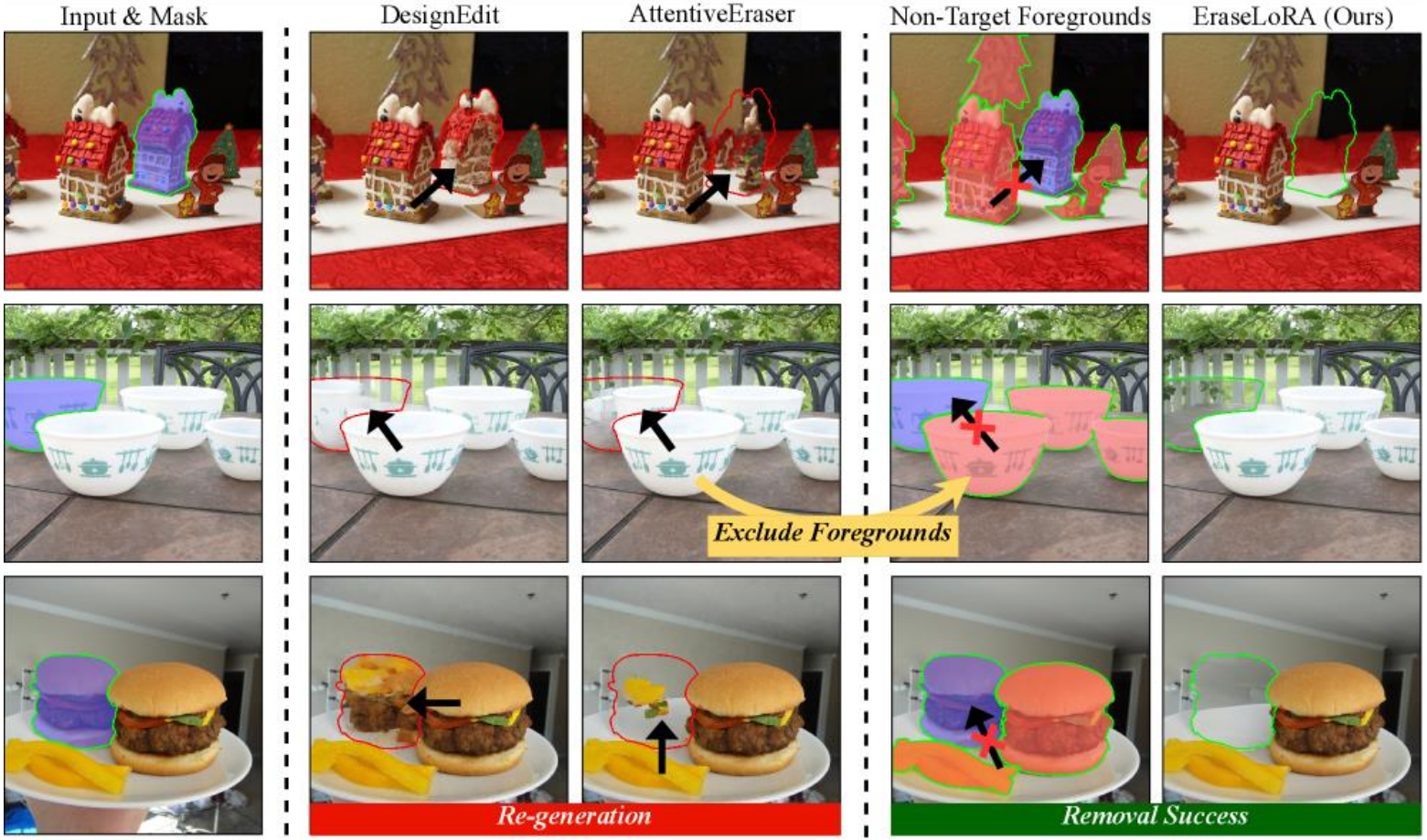


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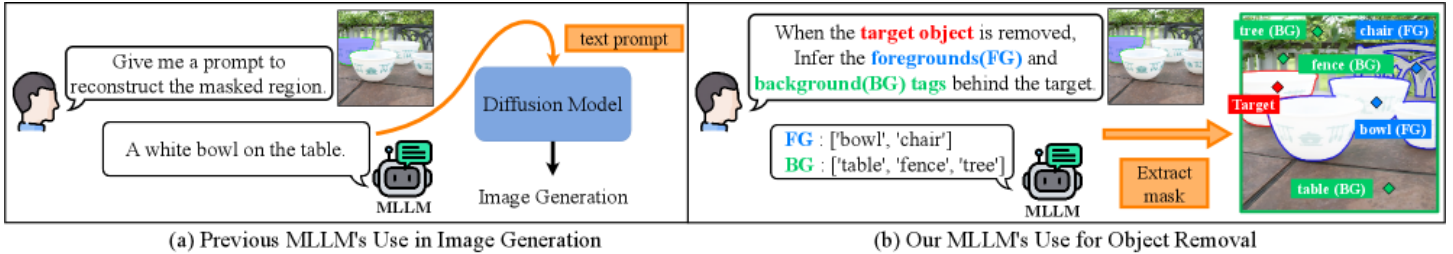
Method	Multi-Class Accuracy (↑)					Multi-Instance Accuracy (↑)					Efficiency (↓)	
	#2	#3	#4	#5	Avg.	#2	#3	#4	#5	Avg.	Latency	VRAM
Qwen-Image [78]	91%	45%	33%	10%	48%	98%	92%	84%	70%	86%	**140s**	**60.1GB**
+ **ISAC$_{LS}$ (Ours)**	**99%**	**58%**	**42%**	**25%**	**56%**	**99%**	**96%**	**89%**	**78%**	**91%**	210s	65.3GB
Flux.2-dev [42]	97%	95%	84%	78%	88%	**100%**	93%	81%	75%	87%	**205s**	**74.2GB**
+ **ISAC$_{LS}$ (Ours)**	**99%**	**98%**	**89%**	**83%**	**92%**	**100%**	**98%**	**88%**	**81%**	**92%**	305s	79.8GB
GPT-Image-1.5 [56]	99%	99%	98%	95%	97%	100%	100%	99%	94%	98%	N/A	N/A
Nano Banana 2 [23]	99%	97%	93%	92%	95%	100%	100%	100%	95%	99%	N/A	N/A



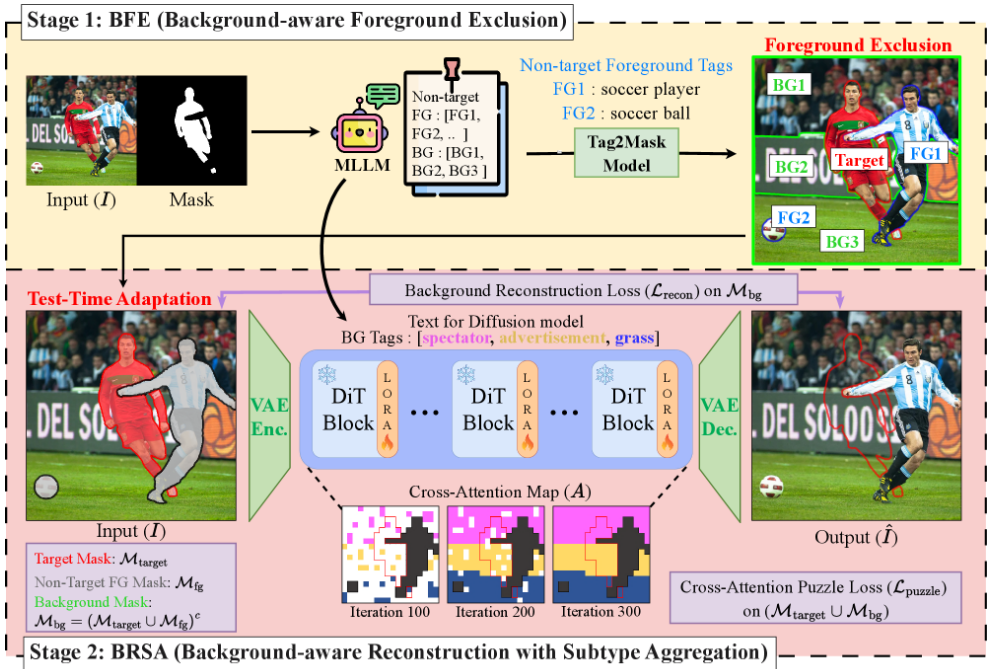
[Under Review] EraseLoRA: Erasing Objects Without a Dataset



[Under Review] EraseLoRA: Erasing Objects Without a Dataset



(a) Previous MLLM's Use in Image Generation (b) Our MLLM's Use for Object Removal

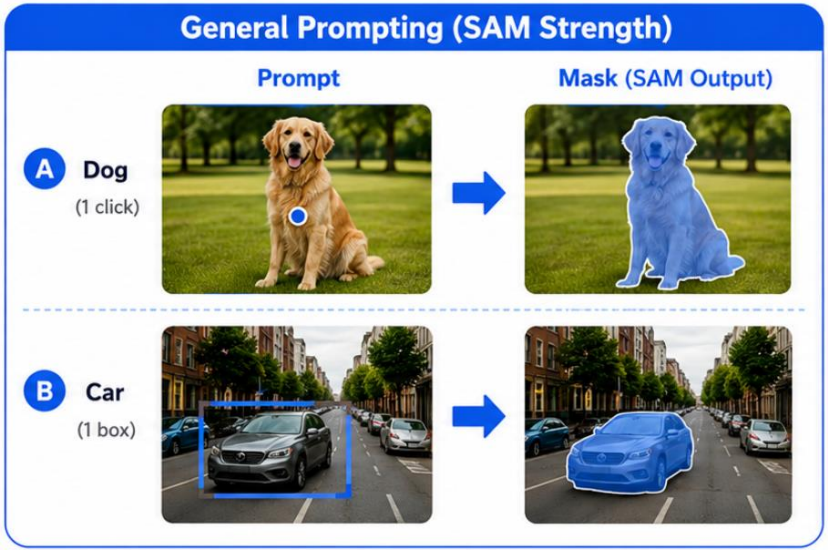


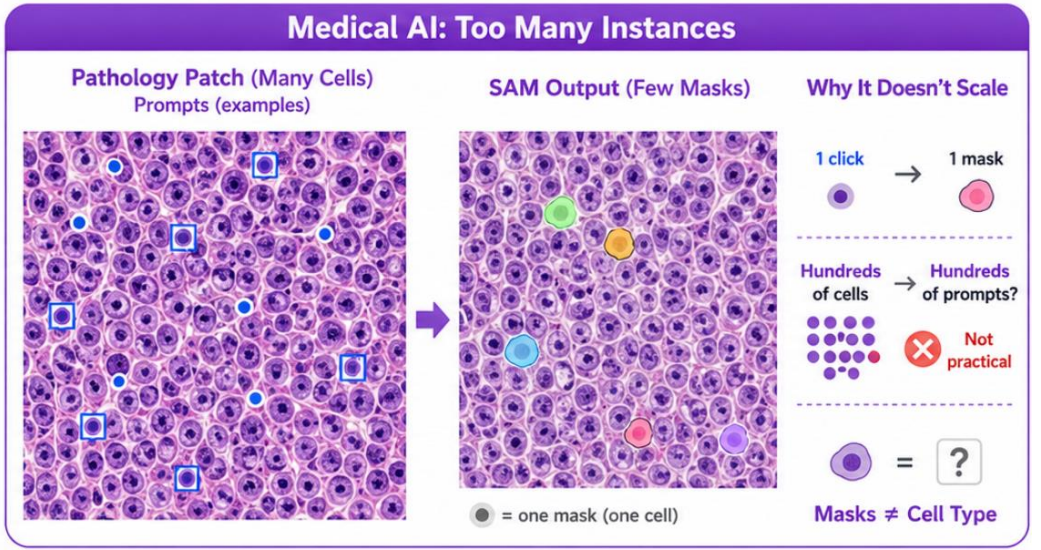
[Under Review] D2R / SOTA Object Composition



What is SAM?

- Segment Anything Model segments almost any object from a simple prompt like a click or a box.

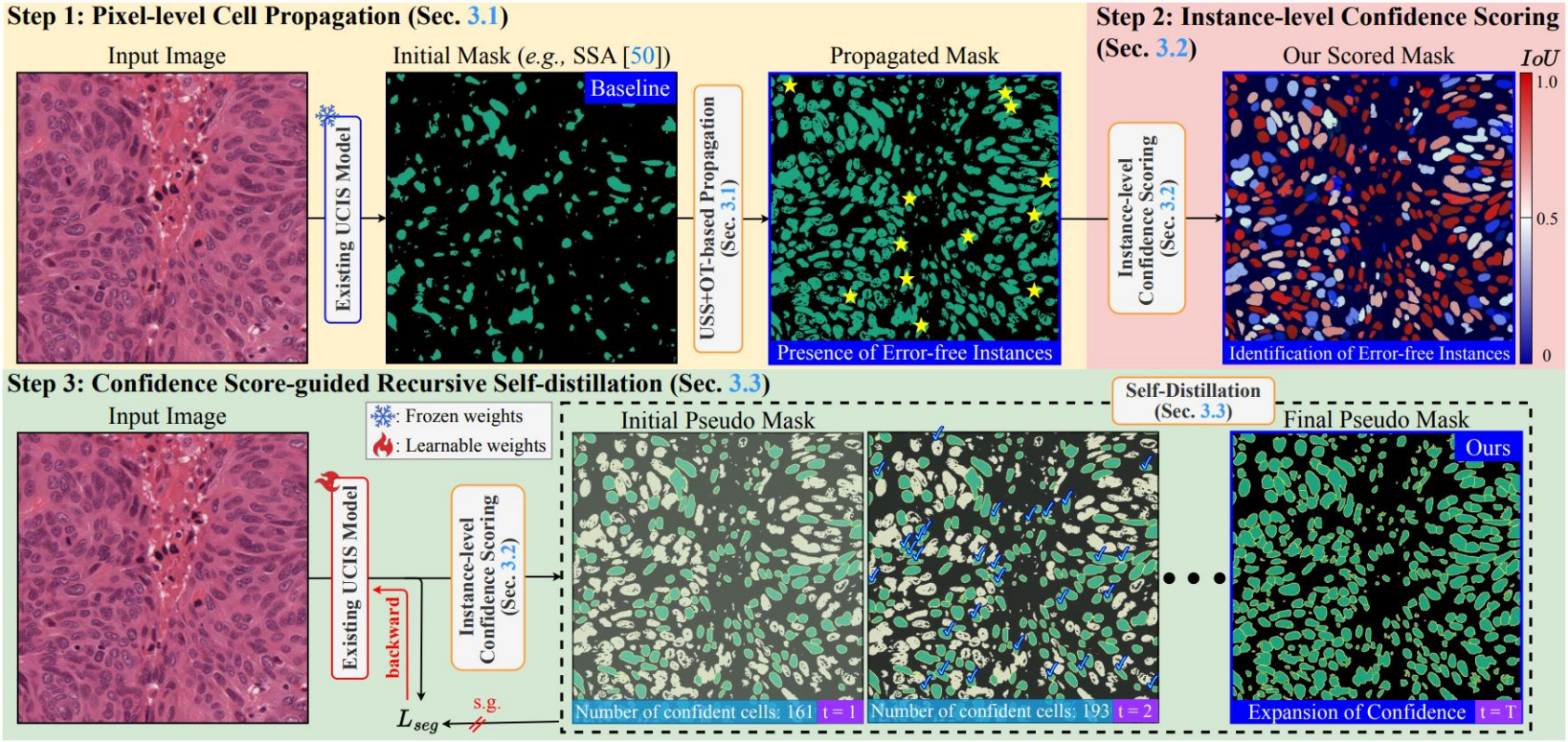






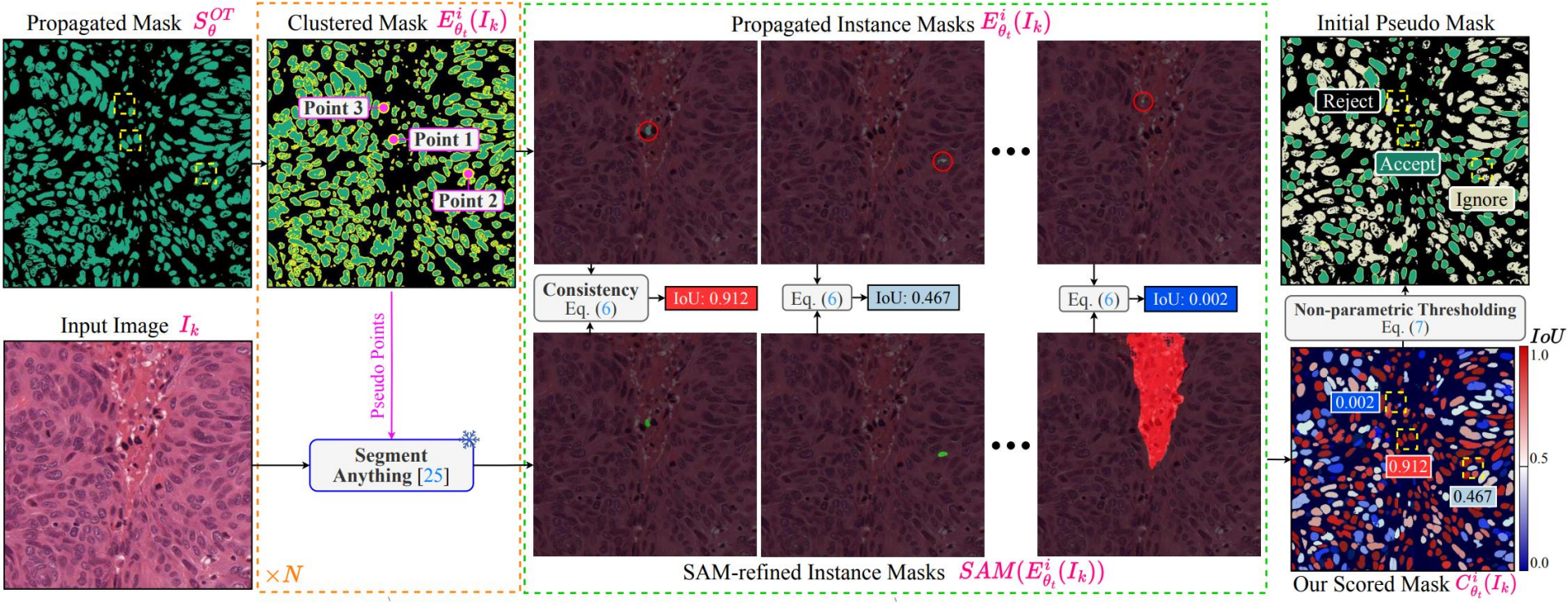
[ICCV 2025] COIN: Segmenting Cells Without Annotations

- COIN teaches a model to segment cells with no human labels.
- The foundation for bringing label-efficient vision into pathology.



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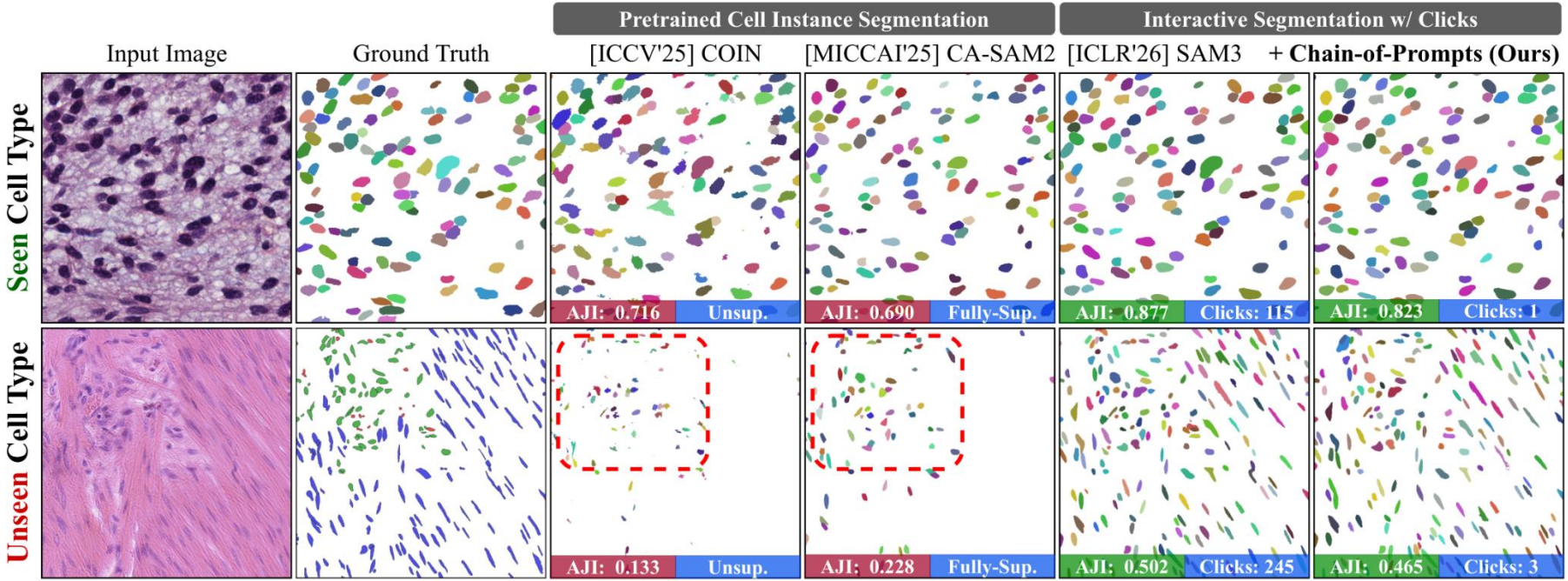
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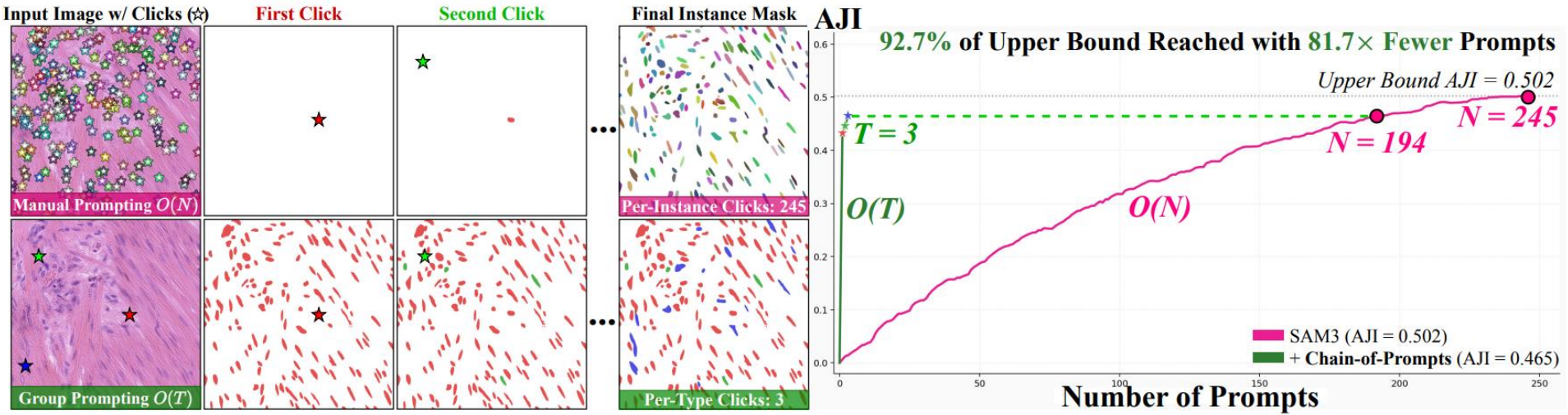
Method	Cell Supervision	MoNuSeg				TNBC			
		AJI (↑)	PQ (↑)	IoU (↑)	Dice (↑)	AJI (↑)	PQ (↑)	IoU (↑)	Dice (↑)
Annotation-free Instance Segmentation									
MaskCut [59] CVPR'23	✗	0.001*	0.000*	0.053*	0.089*	0.006*	0.000*	0.048*	0.088*
CutLER [59] CVPR'23	✗	0.002*	0.000*	0.143*	0.244*	0.003*	0.000*	0.082*	0.146*
ProMerge [32] ECCV'24	✗	0.000*	0.000*	0.013*	0.024*	0.004*	0.000*	0.046*	0.076*
Annotation-free Cell Instance Segmentation									
CellProfiler [4] Genome Biology'06	✗	0.123	-	-	0.404	0.208	-	-	0.415
Fiji [51] Nature Methods'12	✗	0.273	-	-	0.665	-	-	-	-
Hou *et al.* [18] CVPR'19	✗	0.498	-	-	0.750	-	-	-	-
SSA [50] MICCAI'20	✗	0.259*	0.185*	0.618*	0.575*	0.273*	0.253*	0.647*	0.538*
SSA + COIN (Ours)	✗	**0.580**	**0.536**	**0.776**	**0.794**	**0.568**	**0.540**	**0.797**	**0.774**
PSM [5] MICCAI'23	✗	0.471*	0.355*	0.689*	0.682*	-	-	-	-
PSM + COIN (Ours)	✗	**0.579**	**0.539**	**0.777**	**0.797**	-	-	-	-
Weakly-supervised Cell Instance Segmentation									
Qu *et al.* [46] MIDL'19	Point	0.496	-	-	0.702	-	-	-	-
C2FNet [53] MICCAI'20	Point	0.493	-	0.624	-	-	-	-	-
Mixed Anno [47] ISBI'20	Point & Mask	0.516	-	-	0.733	-	-	-	-
BB-WSIS [58] MICCAI'21	Box	-	-	-	0.728	-	-	-	0.703
Liu *et al.* [35] ISBI'22	Point	0.534	-	-	0.740	-	-	-	-
SPPNet [63] MLMI'23	Point	0.497*	0.392*	0.709*	0.719*	-	-	-	-
All-in-SAM [8] IOPscience'23	Box	0.502	-	-	0.738	-	-	-	-
PROnet [38] MICCAI'23	Point	0.555	-	-	0.750	-	-	-	-
InstaSAM [39] MICCAI'24	Point	0.574	-	-	0.772	-	-	-	-
Semi-supervised Cell Instance Segmentation									
CDCL [61] CVPR'22	Mask	-	-	-	0.782	-	-	-	-
TextDiff [11] MICCAI'24	Mask & Text	0.510*	0.410*	0.726*	0.726*	0.464*	0.358*	0.728*	0.666*

[ACCV 2026 Oral] One Click per Cell Type Suffices

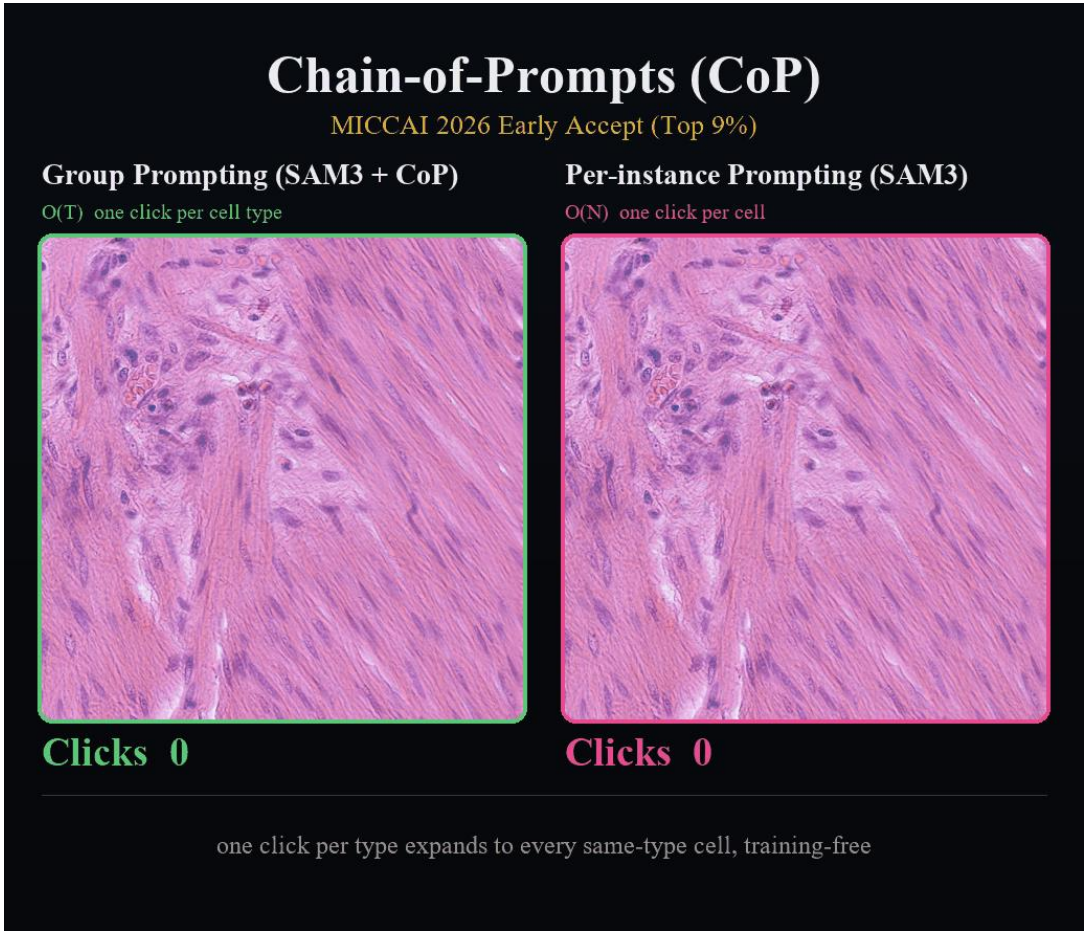
- A pathologist gives a single click per cell type. Training-free group interaction turns it into dense cell instance masks.
- The label-efficient idea completed. We evolved from tags to zero labels and finally to one click.



[ACCV 2026 Oral] One Click per Cell Type Suffices



[ACCV 2026 Oral] One Click per Cell Type Suffices



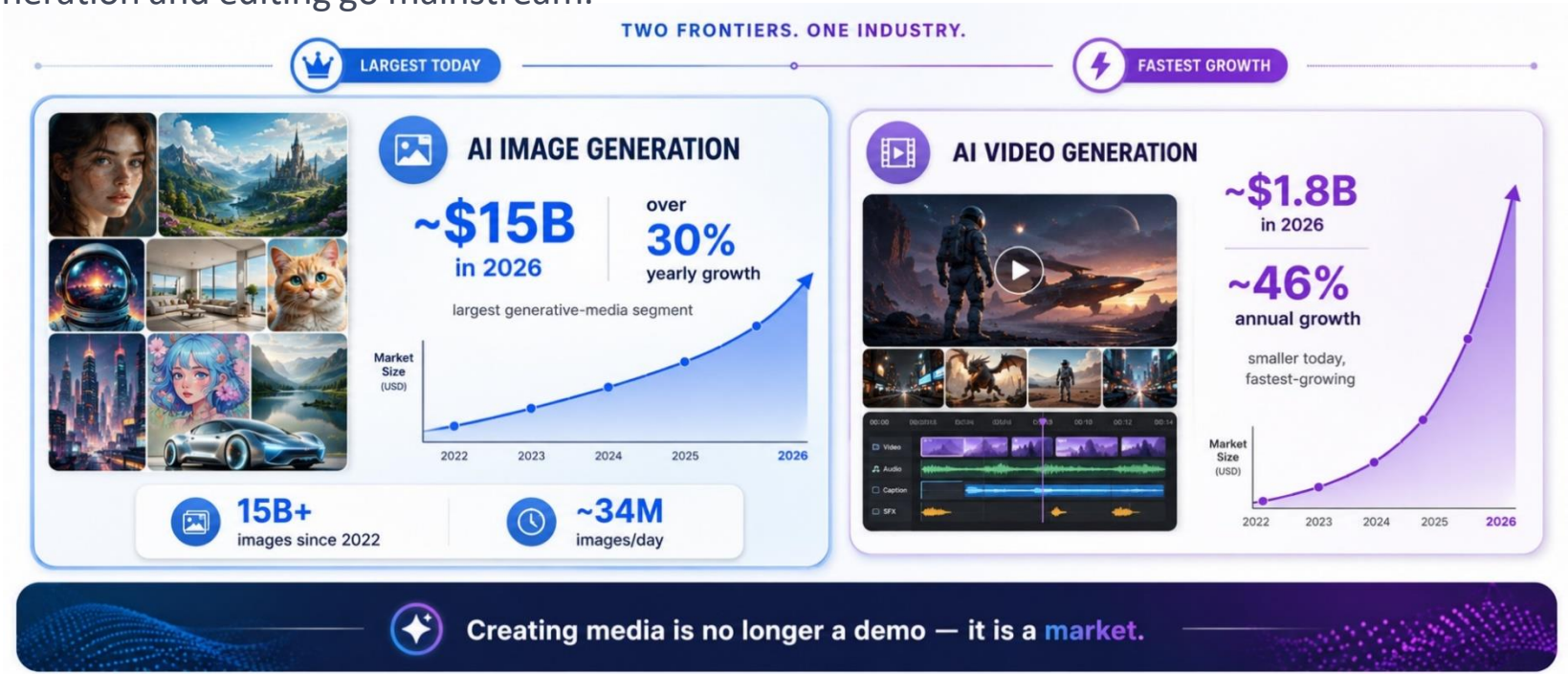
The Generator Steps Outside the Lab

- Having just explored the hidden workings of generative models, we now see these same models already redrawing entire industries out in the real world.
- Vision AI is changing the world along two key frontiers: creating media and modeling reality itself.



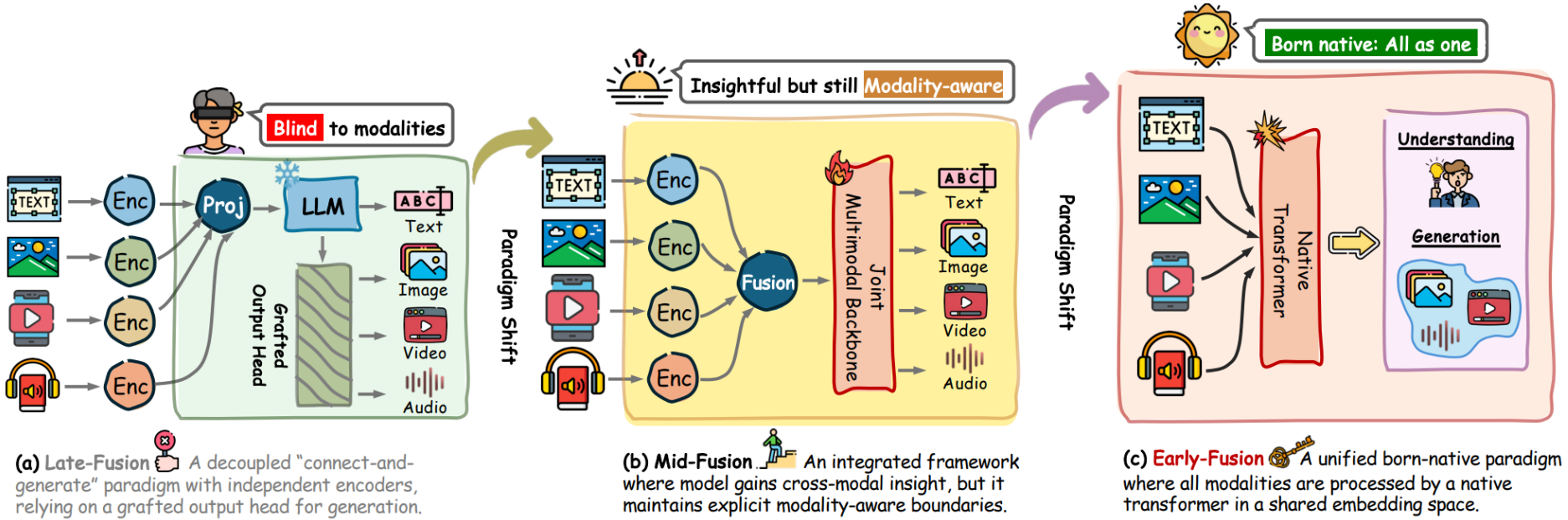
Generating Reality is Now an Industry

- AI image generation is already the largest generative-media segment, around USD 15 billion in 2026 and growing over 30% a year.
- AI video is smaller but fastest-growing, multiplying several times this decade as photorealistic generation and editing go mainstream.



How One Model Does It All

- The old way bolted a vision encoder onto a language model; the backbone stayed blind to raw pixels.
- A single transformer handles every modality in one shared space from the start, a born-native model.



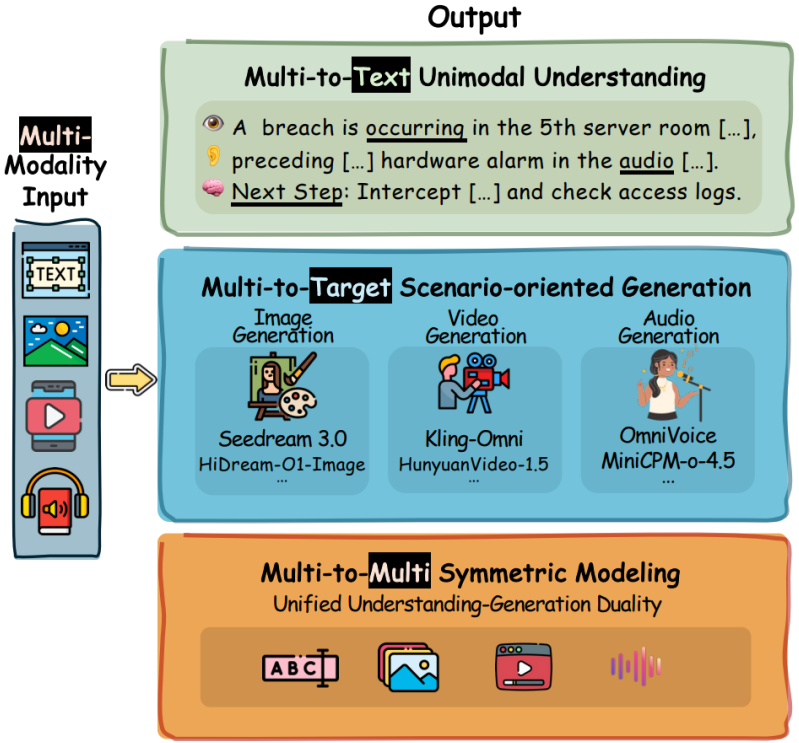
(a) Late-Fusion A decoupled "connect-and-generate" paradigm with independent encoders, relying on a grafted output head for generation.

(b) Mid-Fusion An integrated framework where model gains cross-modal insight, but it maintains explicit modality-aware boundaries.

(c) Early-Fusion A unified born-native paradigm where all modalities are processed by a native transformer in a shared embedding space.

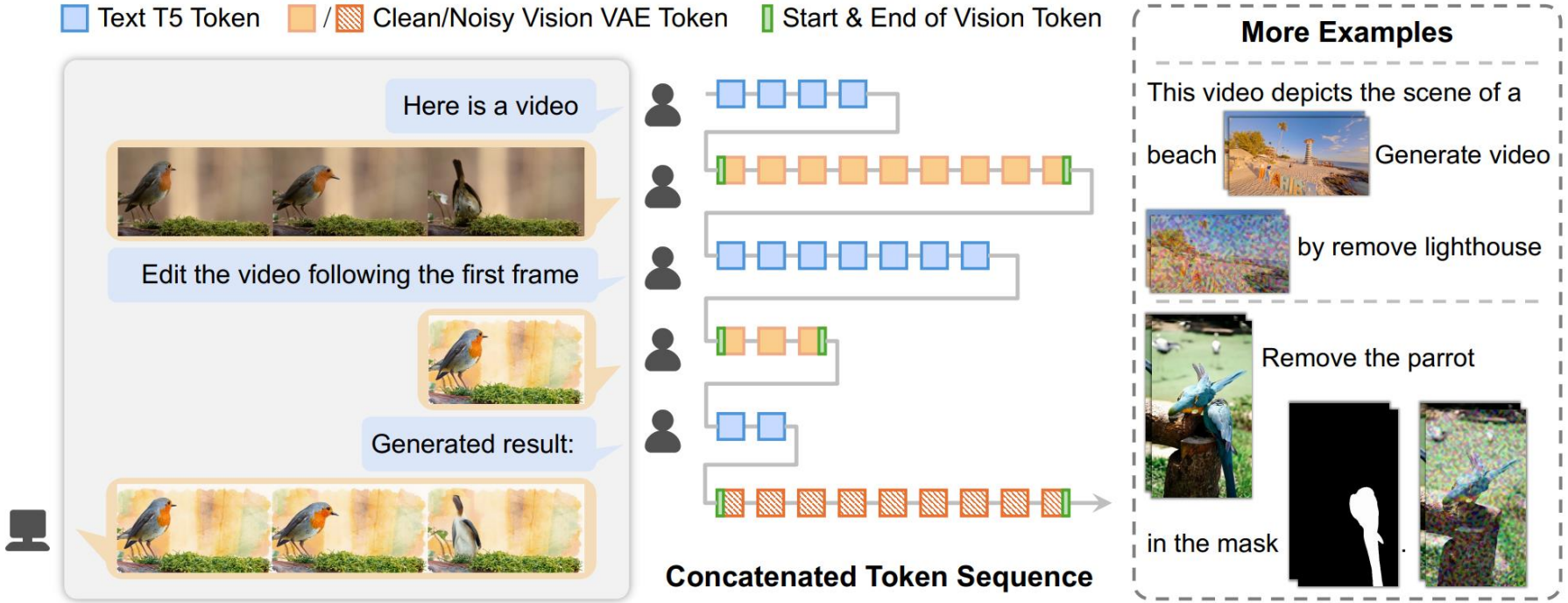
Understanding and Generation as One

- **Three regimes:** read inputs into text (M2T), generate a target like image or video (M2G), or do both symmetrically (M2M).
- The endpoint of M2M is a single network where understanding and generation coexist.



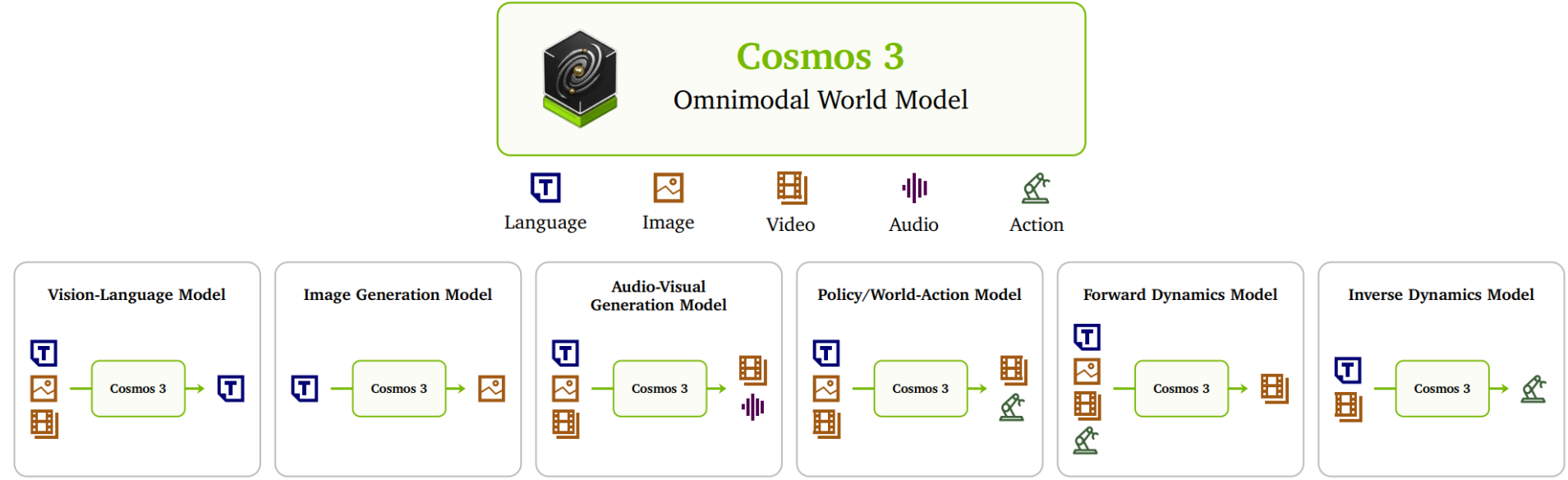
Understanding and Generation as One

- **[ICLR 2026 Oral]** EditVerse: Unifying Image and Video Editing and Generation with In-Context Learning

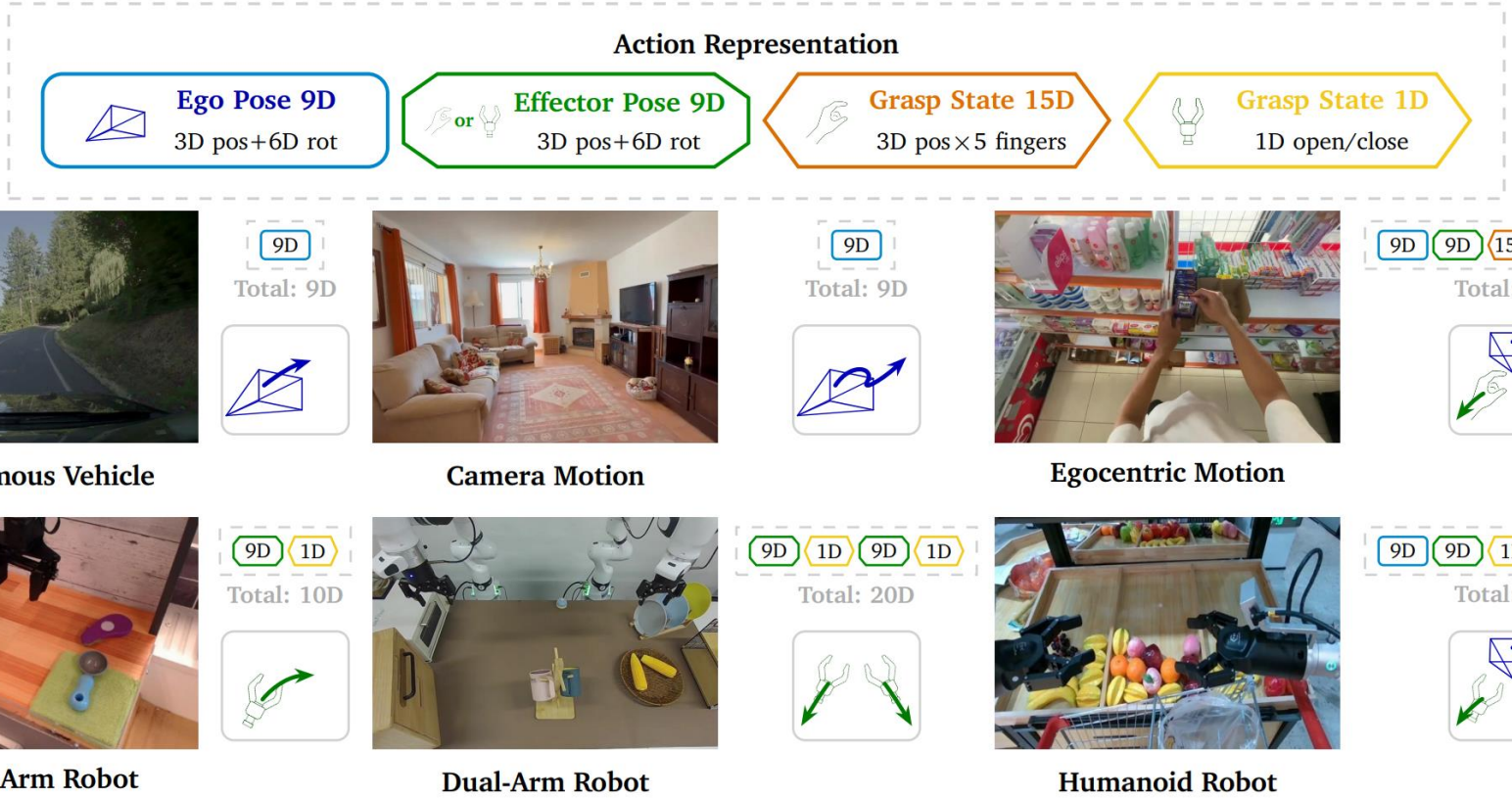


From Images to Worlds

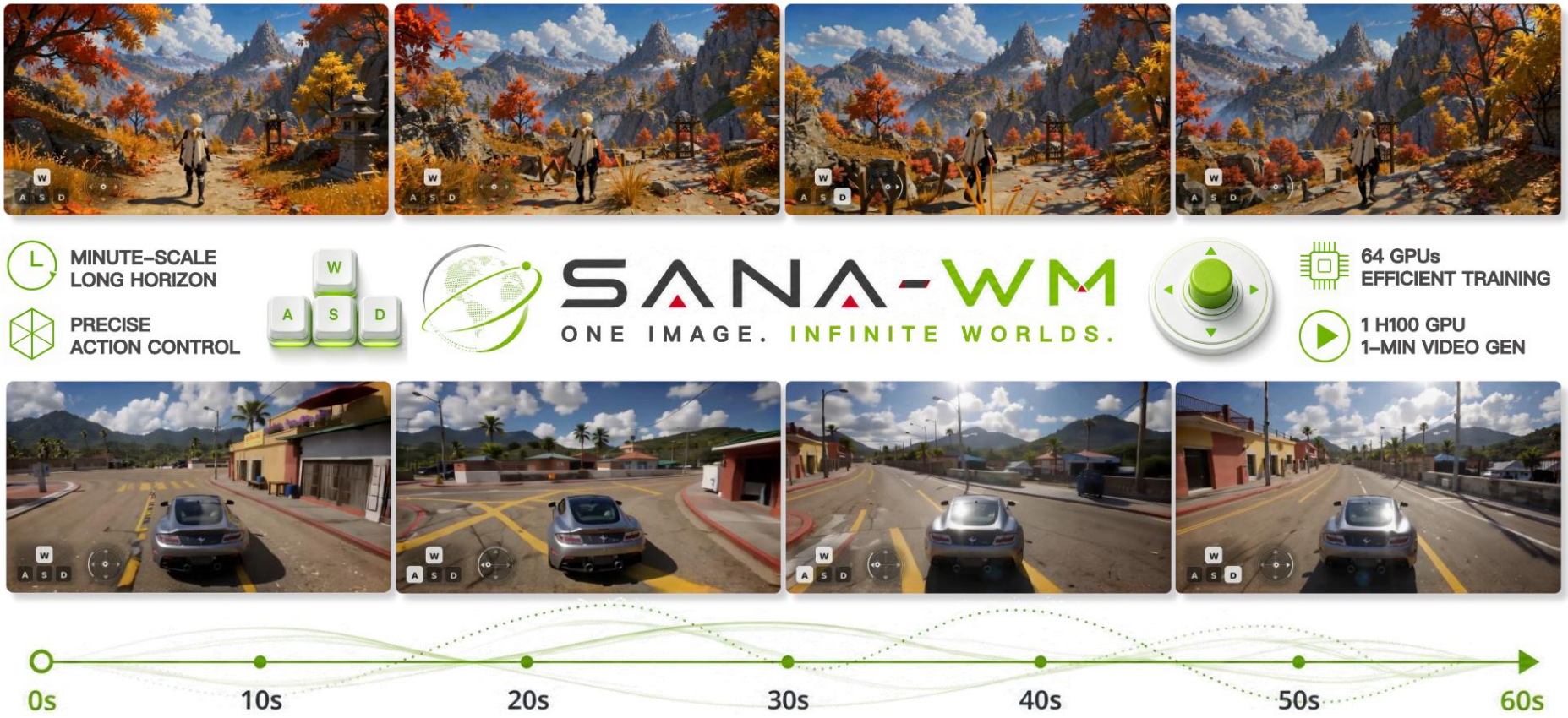
- Beyond single images and clips, world models generate physically plausible worlds and predict what happens next.
- They train robots and self-driving cars in simulation, cutting development from months to days.



From Images to Worlds

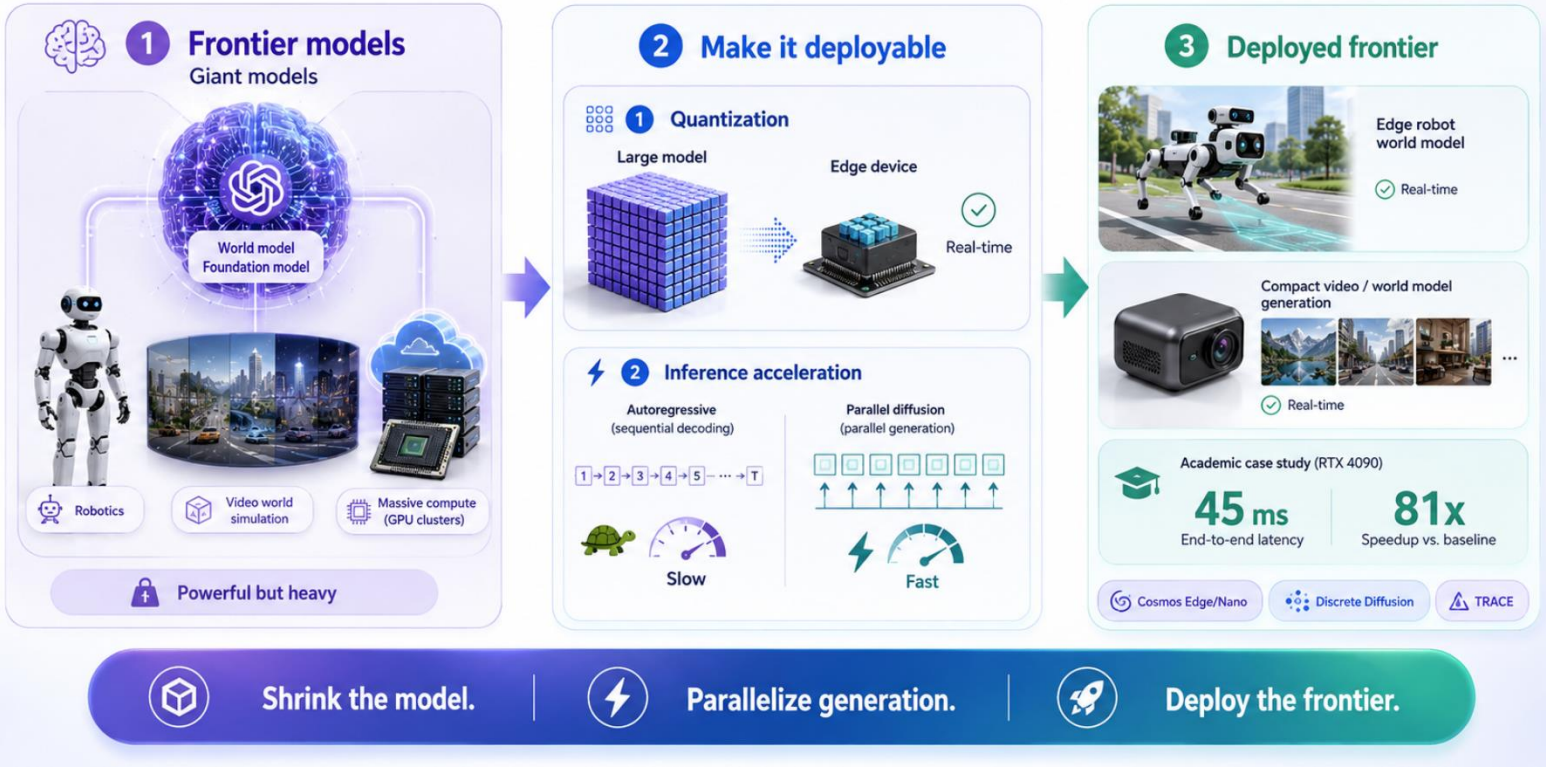


From Images to Worlds



Making the Frontier Deployable

- Quantization shrinks giant models for edge devices, bringing world models to real-time inference.
- Moving generation from slow autoregressive decoding toward parallel diffusion for large speedups.



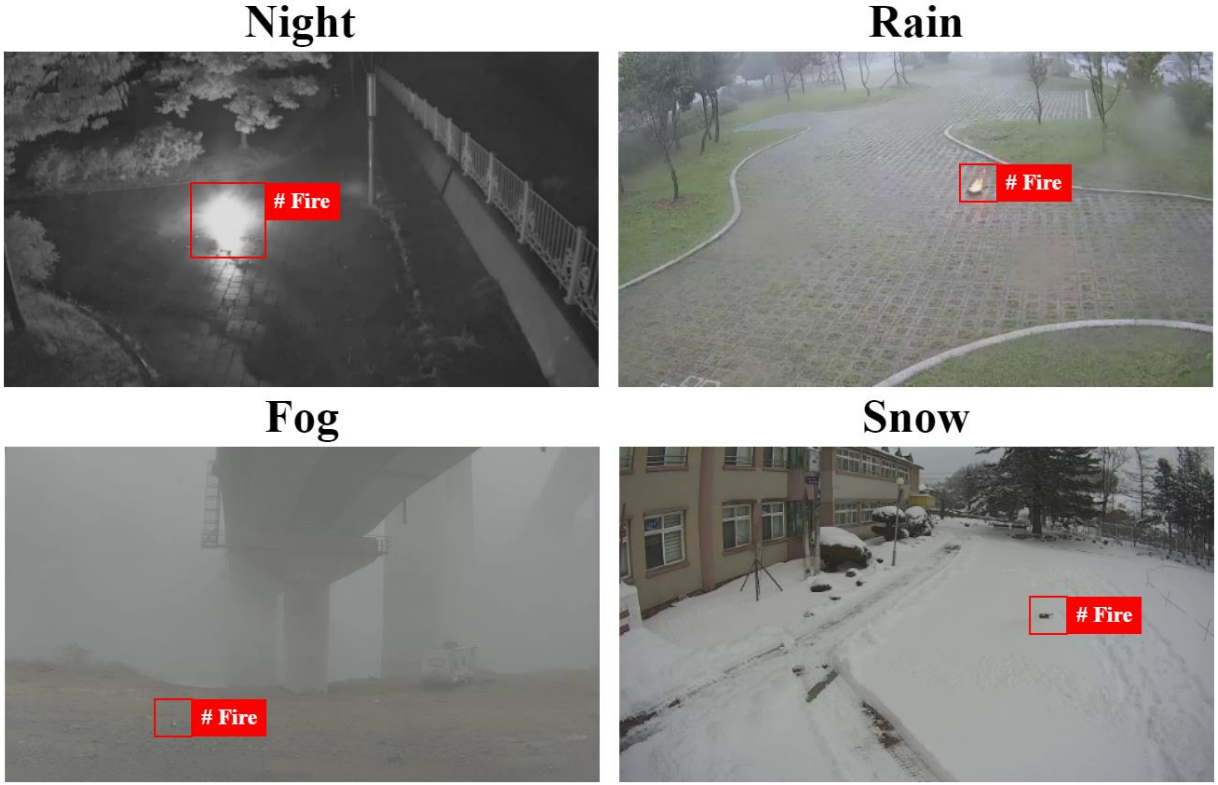
The Common Engine: Doing More with Less

- Media, world models, robots, edge: all advance by needing fewer labels, less compute, less latency.
- Now redrawing the frontier, and the foundation OGQ builds products on next!



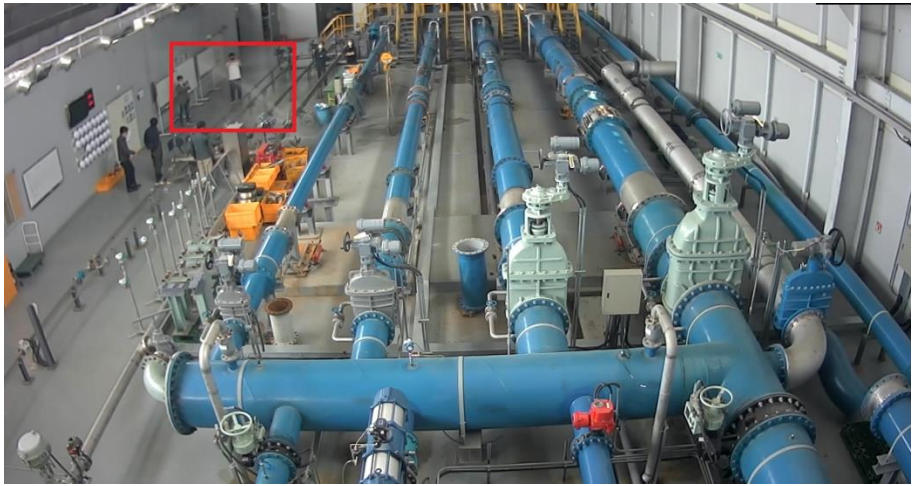
Private & On-Device: Vision for Public Safety

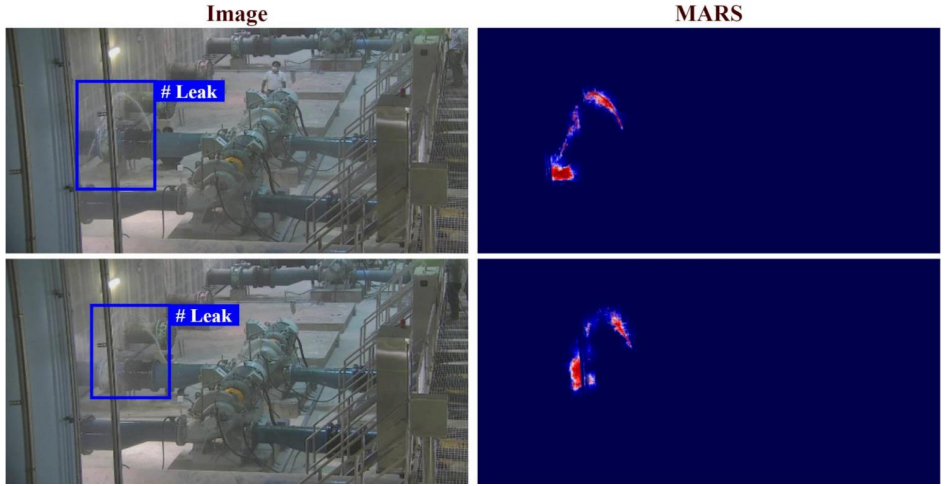
- **Robust Field Detection:** Validated on real-world data to accurately detect leaks, fires, and structural collapses across extreme conditions, including night, rain, fog, and snow.



Private & On-Device: Vision for Public Safety

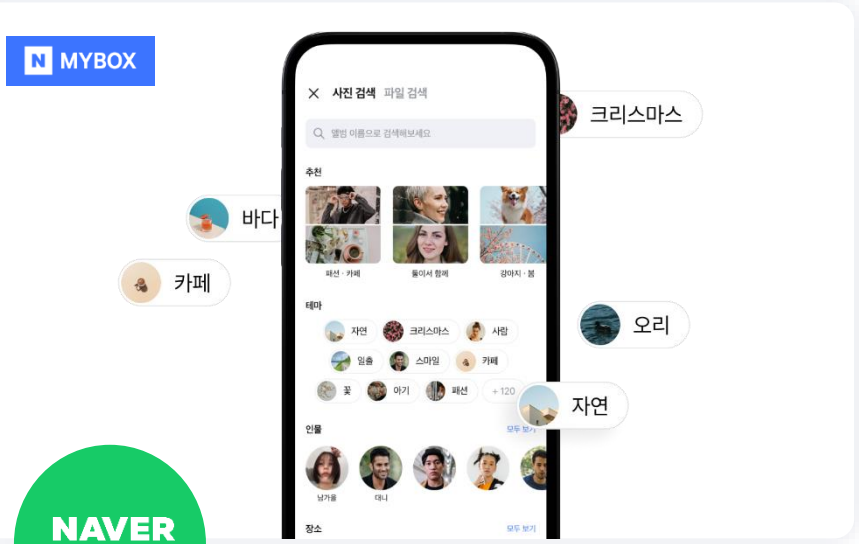
- **Privacy-Preserving & On-Device Deployment:** Rolled out with public partners (K-water, IITP, and police drone programs). Powered by our latest TRACE research to ensure high performance without compromising data privacy.

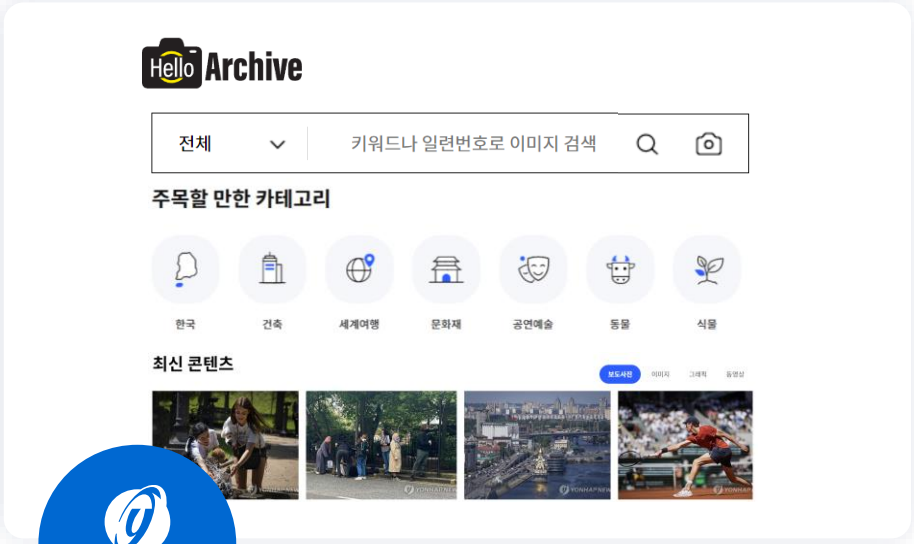




Content Search: Making Assets Findable

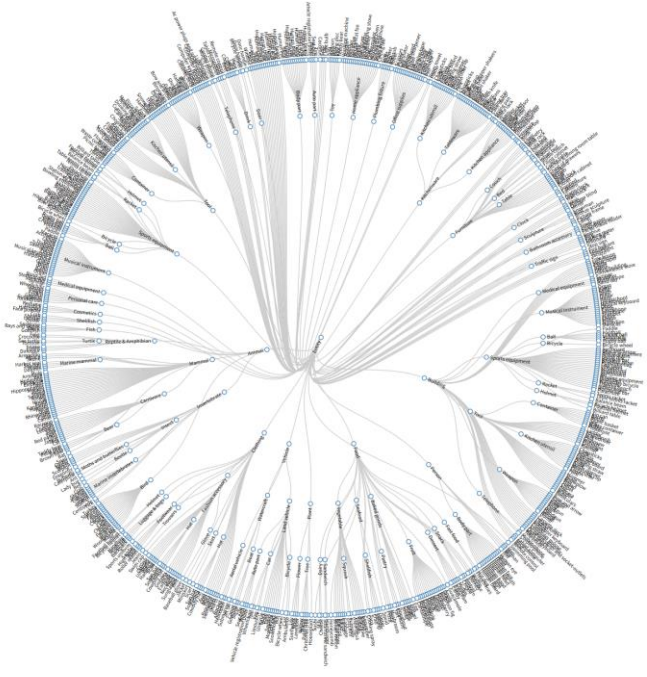
- **Semantic Auto-Tagging:** Transforming massive asset catalogs to be searchable by their underlying meaning, moving beyond traditional filename searches.
- **Powering Partner Ecosystems:** Driving the core content search engine for major platforms like Naver MYBOX, directly leveraging our foundational recognition research.





Content Search: Making Assets Findable

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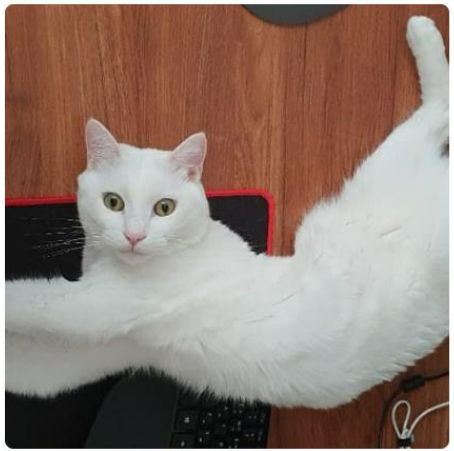


tag_id	en	check	테마명	ko1	ko2
1	plant			식물	공장
2	person	t	사람	사람	
3	sky	t	하늘	하늘	
4	flower	t	꽃	꽃	
5	food	t	음식	음식	
6	tree	t	나무	나무	
7	font	s		폰트	
8	illustration	s		삽화	
9	nature	t	자연	자연	
10	natural landscape	s		자연 경관	
11	cuisine	s		요리법	요리
12	flowering plant	s		꽃식물	꽃 피는 식물
13	dish	t	접시	접시	
14	ingredient			재료	
15	architecture			건축	건축학
16	photography	s		사진술	사진
17	logo	s		로고	
18	graphics	s		삽화	
19	text	t	텍스트	본문	텍스트
20	building	t	건물	건축물	
21	pink	t	분홍색	분홍	분홍색
22	product	t	제품	제품	

Secondary Creation: Editing for Creators

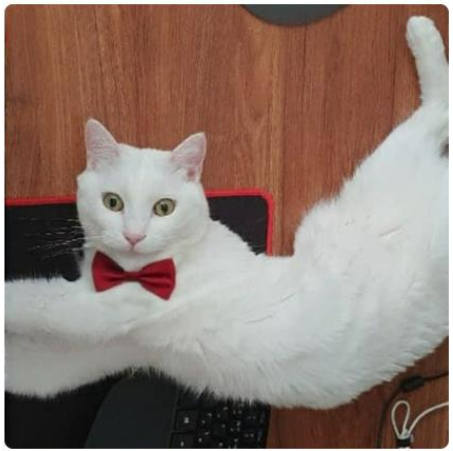


Secondary Creation: Editing for Creators



Original

1.107



Edited (Round 6)

2.143 (+1.04)

Best

2.143

+1.04

CONTENT RATING

ORIGINAL				EDITED			
general	95.8%	sensitive	4.6%	general	95.9%	sensitive	4.4%
questionable	0.1%	explicit	0.0%	questionable	0.2%	explicit	0.0%

DETECTED TAGS

ORIGINAL	EDITED
cat 0.931	bowtie 0.926
animal focus 0.766	bow 0.916
animal 0.761	cat 0.905
white cat 0.753	no humans 0.817
looking at viewer 0.735	animal focus 0.803
no humans 0.685	red bow 0.732
solo 0.522	looking at viewer 0.717
white fur 0.484	animal 0.709

the cat is wearing a cute bow tie.

Edit | New Image



Before 4.83



After 6.00



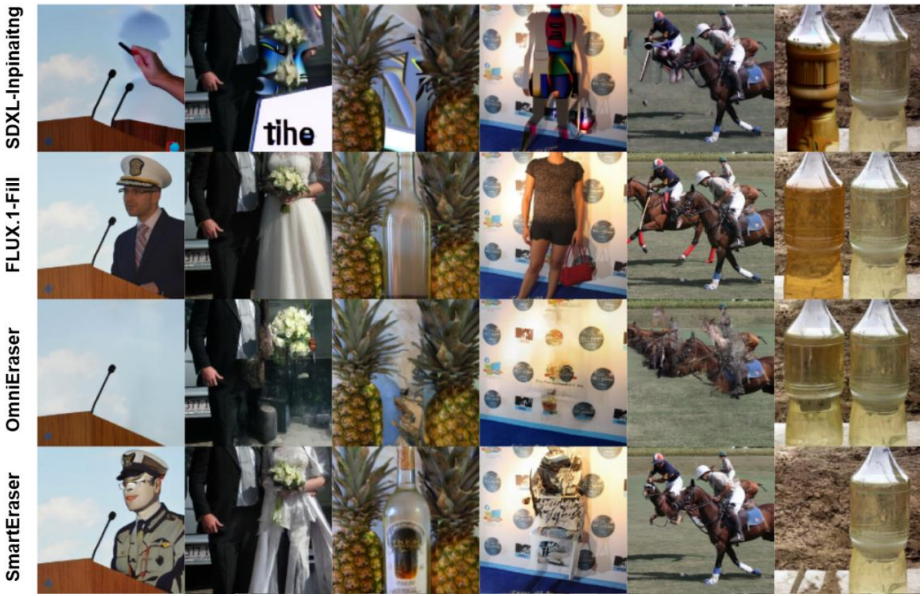
Before 7.85

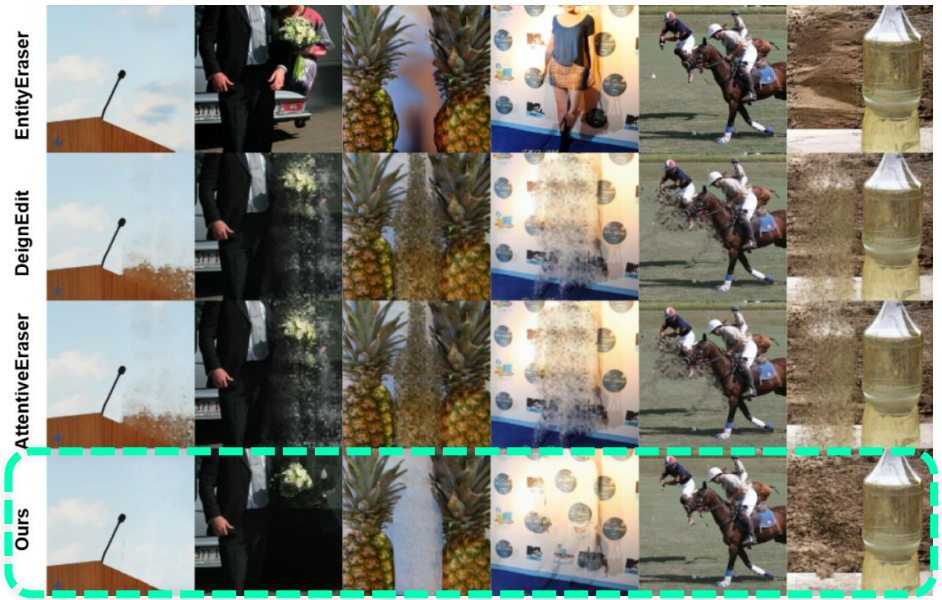


After 8.97

Secondary Creation: Editing for Creators

- **Empowering Derivative Works:** Equipping creators with intuitive prompt-based editing and intelligent object removal tools to accelerate secondary creation.
- **Research-Backed Innovation:** Delivered through national R&D initiatives (NIPA) and built directly upon our core Diffusion and EraseLoRA.





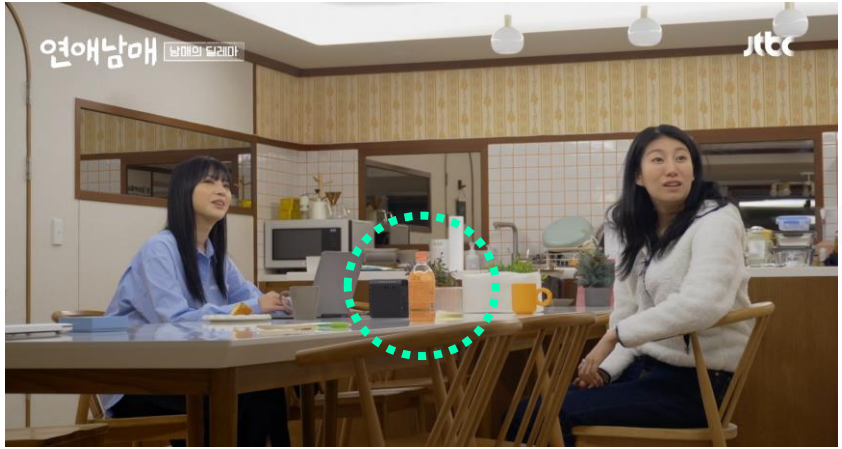
Secondary Creation: Editing for Creators

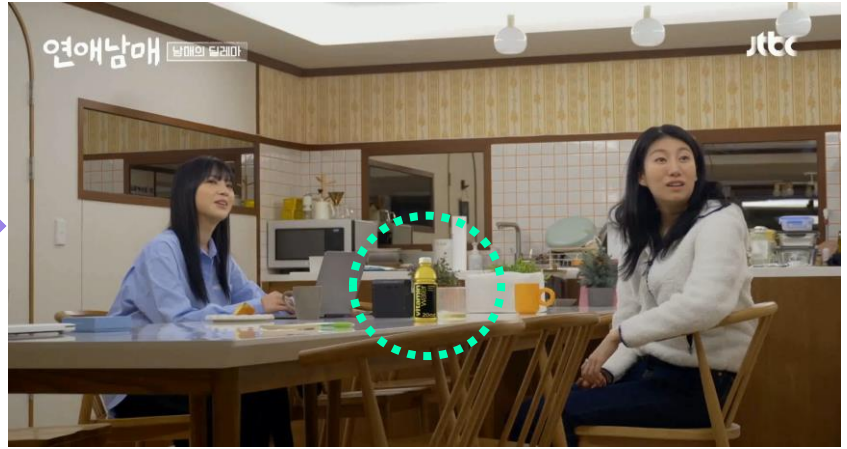
w/o AI PPL



w/ OGQ AI PPL







Secondary Creation: Editing for Creators



Secondary Creation: Editing for Creators

- **ONAIR:** Broadcast footage with overlaid subtitles, captions, and visual effects.
- **CLEAN:** Original footage before post-production overlay.

Input Video



Output Video



The Academia ↔ Industry Flywheel

- **Real-World Constraints as Catalysts:** Industrial challenges, such as labeling costs, deployment speeds, and on-device limits, directly fuel our research agenda.
- **The Continuous Innovation Loop:** These practical challenges inspire top-tier academic papers like TRACE and CoP, which immediately flow back to upgrade our core products.

